Microscopic Derivation of Ginzburg Landau “like” theory for High Tc Superconductors using Extremely Correlated Systems Approach

A thesis submitted

in partial fulfillment for the award of the degree of

Master of Science

in

Solid State Physics

by

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March 2025

Certificate

This is to certify that the thesis titled ***Microscopic Derivation of Ginzburg Landau "like" theory for High Tc Superconductors using Extremely Correlated Systems Approach*** submitted by **Apoorv Srivastava**, to the Indian Institute of Space Science and Technology, Thiruvananthapuram, in partial fulfillment for the award of the degree of **Master of Science** in **Solid State Physics** is a bona fide record of the original work carried out by him/her under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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I declare that this thesis titled ***Microscopic Derivation of Ginzburg Landau "like" theory for High Tc Superconductors using Extremely Correlated Systems Approach*** submitted in partial fulfillment for the award of the degree of **Master of Science** in **Solid State Physics** is a record of the original work carried out by me under the supervision of **Prof. NS Vidhyadhiraja**, and has not formed the basis for the award of any degree, diploma, associateship, fellowship, or other titles in this or any other Institution or University of higher learning. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

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This thesis is dedicated to my family, friends and mentors

Acknowledgements

I am deeply indebted to Dr. N.S. Vidhyadhiraja for introducing me to this invaluable opportunity to work with Prof. T.V. Ramakrishnan and for trusting me with such a prestigious collaboration. His unwavering confidence in my abilities and initial encouragement gave me the motivation to take on this challenging endeavor. His consistent support and mentorship throughout the project have been indispensable.

I also extend my heartfelt thanks to Dr. S.R. Hassan, whose expertise and keen insights have significantly shaped the trajectory of my research. His thoughtful feedback, coupled with his ability to provide clarity on intricate concepts, has greatly enhanced the depth of my work. His enthusiasm and encouragement have been a constant source of inspiration.

I would like to express my deepest gratitude to my Prof. T.V. Ramakrishnan for trusting me with this ambitious project. His guidance and mentorship have profoundly shaped my understanding of condensed matter physics and my approach to research. His ability to connect theoretical insights with experimental realities and his emphasis on physically intuitive models have been an inspiration throughout this thesis work.

A special thanks to Prof. H R Krishnamurthy(ICTS/IISc) and Prof. Mukul Laad (IMSc) for meaningful and insightful discussions.

I would also like to express my appreciation to the Strongly Correlated Electron Systems Lab at JNCASR, where I had the opportunity to explore challenging problems in condensed matter physics, and to Mr. Gopal Prakash at IMSc, whose insightful discussions and help with the development of self consistency codes are invaluable.

I am sincerely grateful to the JNCASR and IMSc for providind funding and facilities during my stays there.

Finally, I wish to thank my peers, collaborators, friends, and family for their unwavering support, which has made this journey possible.

Finally, I dedicate this thesis to everyone who has contributed to my academic and personal growth, shaping me into the researcher I am today.

Apoorv Srivastava

Abstract

We derive micoscopically a Ginzburg Landau (GL) like theory for strongly correlated high T_c superconductors. Working in the faithful Hubbard X operator representation, we start from the Extremely Correlated Fermi Liquid or $U = \infty$ limit of Shastry [1] and use a Schrieffer–Wolff transformation to obtain the effective low energy Hamiltonian upto order $(1/U)$ in this representation. The relevant small term J (which is of order $(1/U)$) describes intersite pair attraction. We then use the Hubbard Stratonovich transformation followed by a cumulant expansion. This gives us the final free energy functional in terms of the order parameter (average spin-singlet nearest-neighbor Cooper pair amplitude) and the GL coefficients in terms of single-particle retarded Green's functions in the $d = \infty$ limit. These are calculated using the theory of Hassan et al. [2]. This functional is used to confront some features of cuprate superconductivity, viz. d-wave superconductive order, the superconducting phase diagram in the hole doping x and temperature plane - the pseudogap curve $T^*(x)$ and the superconducting dome $T_c(x)$.

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Chapter 1

The High Tc Problem

1.1 High-T_c Superconductors

The discovery of high-temperature (high-T_c) superconductors(HTSC) in cuprate materials ($La_{2-x}Ba_xCuO_4$) by Bednorz and Müller in 1986 dramatically transformed the field of condensed matter physics, introducing new theoretical and experimental challenges [4] due to the interplay of strong electron correlations, low-dimensionality, and unconventional pairing mechanisms. Unlike conventional superconductors explained successfully by the Bardeen-Cooper-Schrieffer (BCS) theory, these high-T_c cuprates cannot be adequately described by phonon-mediated pairing mechanisms. Instead, they present complex behaviors rooted fundamentally in electron-electron interactions, placing them within the broader class of strongly correlated electron systems (SCES) [5].

Strongly correlated electron systems represent materials where electronic interactions play an essential and dominant role, overriding predictions made by conventional band theory. Due to significant Coulomb interactions among electrons, SCES exhibit rich and intricate phenomena, including metal-insulator transitions (MIT), unconventional magnetism, and superconductivity [6]. The prototypical example of strong correlation is manifested through the Hubbard model or its strong-coupling limit, the $t - J$ model, where electrons strongly repel each other when occupying the same site, introducing competition between kinetic and potential energy. At half-filling (the undoped state, the parent compound (e.g., La_2CuO_4 or $YBa_2Cu_3O_6$)), electron density corresponds precisely to one electron per lattice site, and strong Coulomb repulsion leads to a gap in the electronic excitation spectrum, resulting in a Mott insulating state. The MIT at half-filling, known as the Mott transition, represents a cornerstone concept in SCES theory and has been extensively explored both theoretically and experimentally [7].

At half-filling, strongly correlated systems typically show insulating behavior because

Coulomb energy outweighs kinetic energy, effectively localizing electrons. Deviations from half-filling, however, introduce charge carriers into the insulating system, substantially altering its electronic properties. This doping procedure transforms insulating antiferromagnets into unconventional metallic phases and, in many cases, can trigger novel forms of superconductivity. For instance, doping a Mott insulator in cuprates gradually suppresses antiferromagnetic ordering and leads to the emergence of a high-T_c superconducting state at specific doping levels [8]. The complexity of these transitions and the competition among various quantum orders has resulted in intense debate and detailed experimental scrutiny.

High-T_c superconductors, particularly cuprates, exemplify how SCES behavior unfolds when transitioning away from half-filling. The superconducting phase emerges from a strongly correlated metallic state, where electron pairing mechanisms deviate significantly from phonon-based BCS superconductivity. These materials typically possess layered, quasi-two-dimensional structures with copper-oxide planes where electronic correlations strongly dominate, resulting in intertwined orders such as superconductivity, antiferromagnetism, charge density waves, and pseudogap phenomena. This rich landscape complicates theoretical modeling significantly [**?**]. In fact the most prominent challenges in HTSCs is understanding the complex doping-temperature (x-T) phase diagram.

One central theoretical challenge in high-T_c superconductors lies in understanding the pseudogap phase that arises when holes are introduced into the system ($x > 0$). It is characterized by a partial gap in the electronic density of states above the superconducting transition temperature. Experimental techniques such as angle-resolved photoemission spectroscopy (ARPES) and scanning tunneling microscopy (STM) have extensively mapped this enigmatic phase, yet its precise microscopic origin remains elusive. Multiple competing explanations exist, including preformed pairs without long-range coherence, short-range antiferromagnetic fluctuations, and various forms of competing orders such as charge-density waves (CDW) or spin-density waves (SDW), or some other exotic phase. The pseudogap phase reflects the inherent complexity emerging from strong electron correlations and is widely viewed as a key barrier in fully comprehending high-T_c superconductivity [5].

As doping increases, superconductivity emerges with a d-wave pairing symmetry, as confirmed by phase-sensitive experiments [9]. The superconducting dome in the x-T phase diagram peaks at optimal doping ($x \approx 0.16$), where the transition temperature T_c reaches its maximum. Beyond this, in the *overdoped regime*, superconductivity weakens, and the system becomes more metallic, approaching a conventional Fermi-liquid behavior. However, even in this regime, the effects of strong correlations persist, leading to deviations from the standard Landau Fermi-liquid picture.

The strange metal phase, observed in the optimally doped to overdoped regions, is another hallmark of HTSCs. Here, the resistivity shows a linear dependence on temperature ($\rho \propto T$) over a wide range, which cannot be explained by conventional electron-phonon scattering mechanisms. This linear resistivity is often associated with a *quantum critical point* hidden beneath the superconducting dome, where the system exhibits critical fluctuations that strongly influence its transport and thermodynamic properties [10]. The strange metal phase highlights the need for theoretical frameworks that go beyond conventional quasiparticle-based descriptions, such as **non-Fermi liquid theories** and approaches grounded in **quantum criticality**.

Theoretical models, including variants of the Hubbard and t-J models, capture the essence of strong electronic correlations but face considerable numerical and analytical difficulties. Advanced computational techniques such as dynamical mean-field theory (DMFT), cluster DMFT, and quantum Monte Carlo simulations have advanced our understanding, providing insights into how electronic correlations lead to unconventional superconductivity. However, these approaches remain computationally intensive and face intrinsic limitations such as the fermionic sign problem, restricting their applicability, especially at realistic parameters relevant to cuprates [7, 11].

Experimental studies have continually emphasized that cuprates are a prime example of SCES, where correlated electron phenomena fundamentally alter their electronic structure and emergent properties. In particular, electronic transport and magnetic susceptibility measurements demonstrate that metallic phases of these materials do not follow conventional Fermi-liquid behavior, instead exhibiting non-Fermi liquid characteristics indicative of strong correlations. The temperature dependence of resistivity often follows nontrivial scaling laws such as linear-in-temperature resistivity, clearly deviating from the T^2-dependence of traditional metals. Such anomalies reinforce the interpretation of cuprates as quintessential strongly correlated systems [12].

The failure of Fermi Liquid (FL) theory in strongly correlated electron systems arises from its reliance on the assumption that the low-energy excitations in the system can be described by well-defined quasiparticles. FL theory works well for weakly interacting electron systems, where these quasiparticles retain a one-to-one correspondence with the underlying particles, and the interactions between them can be treated perturbatively. However, in strongly correlated systems, such as high-temperature superconductors, electron-electron interactions are so strong that quasiparticles are no longer well-defined, and their concept breaks down. As a result, FL theory cannot account for phenomena like the pseudogap phase, the strange metal phase, and the anomalous temperature dependence of resistivity and other transport properties observed in high-T_c superconductors. Thus, the breakdown of FL theory highlights the need for alternative frameworks, such as the ECFL theory, which can handle these strong correlations and the resulting non-Fermi liquid behavior.

1.2 Failure of Fermi Liquid Theory in Strongly Correlated Electron Systems

Fermi liquid theory (FLT), originally proposed by Landau, has been remarkably successful in describing conventional metals where electron-electron interactions are weak or moderate. The central assumption underlying FLT is the adiabatic continuity from a non-interacting electron gas to an interacting electron system. Under this scenario, the interacting electrons can be treated as quasi-particles—electron-like excitations characterized by slightly renormalized masses and finite lifetimes, leading to well-defined Fermi surfaces and quasiparticle states near these surfaces [13]. Mathematically, the self-energy $\Sigma(\mathbf{k}, \omega)$ of electrons near the Fermi surface in FLT has the general low-energy form:

$$\Sigma(\mathbf{k}, \omega) \approx \Sigma(\mathbf{k}_F, 0) + (1 - Z^{-1})\omega - iC(\omega^2), \tag{1.1}$$

where Z is the quasiparticle weight, finite and non-zero for conventional metals, and C is a constant determining quasiparticle lifetime [14].

However, strongly correlated electron systems violate the main assumptions underlying this quasiparticle description due to the very strong electron-electron Coulomb repulsions. In these materials, the electron self-energy $\Sigma(\mathbf{k}, \omega)$ deviates dramatically from FLT predictions, frequently exhibiting non-analytic and singular energy dependence near the Fermi energy. A signature of these non-Fermi-liquid (NFL) states is the vanishing or severe suppression of the quasiparticle residue Z, mathematically expressed as:

$$Z = \left[1 - \frac{\partial \mathrm{Re}\Sigma(\mathbf{k}, \omega)}{\partial \omega}\Big|_{\omega=0}\right]^{-1} \to 0. \tag{1.2}$$

This indicates that the quasiparticle description becomes invalid, reflecting the breakdown of the basic FLT assumption of weak interactions and perturbative continuity from free-electron models [15].

Experimentally, this results in anomalous physical properties, such as resistivity varying linearly with temperature rather than the FLT-predicted quadratic dependence, and unusual magnetic susceptibilities or thermodynamic responses. Hence, the inherent complexity and strong correlations in such systems fundamentally limit the applicability of FLT, requiring alternative non-perturbative approaches and theories, such as Dynamical Mean Field Theory (DMFT)[7] or sophisticated numerical methods, to capture their unconventional electronic behaviors accurately [11].

1.3 Extremely correlated Fermi Liquid Theory

The *Extremely Correlated Fermi Liquid* (ECFL) theory, developed by B.S. Shastry, provides a powerful framework for understanding systems with strong electron-electron correlations, particularly high-temperature superconductors. In these systems, traditional approaches such as Fermi liquid theory fail due to the breakdown of quasiparticles and the emergence of non-Fermi liquid behavior. The key challenge arises from the projection of the Hilbert space caused by Coulomb repulsion, which forbids double occupancy and leads to complex correlations that cannot be captured by conventional perturbative methods.

He starts by writing the HUbbard Hamiltonian in the $U = \infty$ limit thereby completely forbidding the double occupancy state. This changes the completeness relations since the Hamiltonian in projected to a subspace of the Hamiltonian. Shastry's approach incorporates *X-operators* to explicitly handle the projection of electron states. These operators are crucial because they allow a systematic treatment of the projection effects without resorting to uncontrolled approximations.

$$X_i^{00} + X_i^{\downarrow\downarrow} + X_i^{\uparrow\uparrow} = \mathbb{1}$$

The inclusion of X-operators ensures that the theory can account for the strong local correlations seen in these systems. Unlike Feynman diagrams and Wick's theorem, which fail in the strongly correlated regime due to the absence of well-defined quasiparticles, Shastry uses a more general method based on the Green's function approach, which avoids the assumptions of weak coupling and quasiparticle existence. Feynman diagrams break down in this context because they assume a perturbative expansion around a non-interacting system, which is not valid in highly correlated materials where electron-electron interactions are dominant.

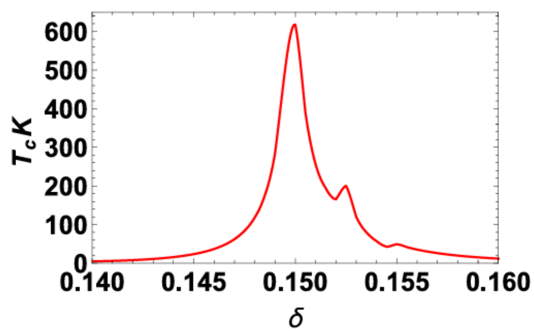
To overcome these challenges, Shastry uses the formalism of the Tomonaga-Schwinger method, which applies to systems with strongly interacting particles. This approach treats the correlation effects non-perturbatively, providing a more accurate description of strongly correlated systems, such as high-temperature superconductors. The ECFL framework relies on this method, enabling the inclusion of the projection effects and allowing for the study of spectral properties, transport coefficients, and superconductivity in a manner that captures the physics of strongly correlated systems.

The theory has been particularly useful in explaining the asymmetric spectral functions observed in cuprates, where experiments show pronounced particle-hole asymmetry. ECFL also sheds light on the mechanisms of high-T_c superconductivity, where pairing is not due to conventional electron-phonon interaction but rather arises from spin fluctuations and strong electron correlations. By using ECFL, Shastry has provided a deeper understanding of the microscopic origins of superconductivity in these systems, offering a controlled method to connect microscopic models to experimentally measurable quantities.

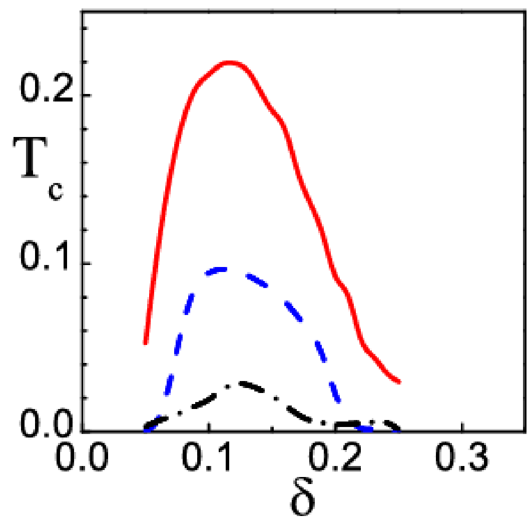
Through his work, Shastry has demonstrated the relevance of the ECFL approach in describing the anomalous normal state behavior and superconductivity of high-temperature superconductors, making it an essential framework for studying extremely correlated electron systems. His contributions highlight the importance of non-perturbative methods and the use of X-operators in handling the complexities of strongly interacting systems [16, 1, 17].

His recent work applies this approach to study superconductivity in $t - J$ Model[18]. Exact equations for the Green's functions are derived by generalizing Gor'kov's formalism to account for the extremely strong local repulsion between electrons with opposite spins. These equations are then expanded in terms of a parameter λ, which quantifies the fraction of double occupancy. At leading order and near the critical temperature T_c, the equations are further simplified (saddle point approximation) to yield an approximate integral equation for the superconducting gap.

In a similar spirit, N. M. Plakida's work [19, 20, 21, 22] extensively utilize advanced Green's function techniques and Hubbard operator formalism, providing substantial insights into the fundamental behaviors observed in high-temperature superconductors (HTSCs), especially cuprates. A prominent feature of Plakida's framework involves explicitly dealing with strong electron-electron interactions through Hubbard operators, effectively capturing the essence of strong correlations inherent in these materials. His method clearly emphasizes the critical role played by spin fluctuations and magnetic excitations in facilitating superconductivity, notably underscoring their relevance as pairing mechanisms in cuprates. Additionally, Plakida's approach systematically incorporates the effects of doping, temperature variations, and magnetic interactions into self-consistent equations, thus allowing accurate reproduction of key experimental observations such as the pseudogap and anomalous superconducting behavior.



(a) Superconducting Dome from Shastry[18]



(b) Superconducting Dome Plakida[22]

Figure 1.1: Superconducting Dome from existing theories using $U = \infty$ paradigmatic limit

The core distinction between Plakida and Shastry lies fundamentally in their methodological approaches. Plakida adopts a generalized Green's function methodology that leverages spin-fluctuation-induced pairing as a central element, explicitly considering magnetic interactions and allowing more phenomenological freedom in adjusting parameters to fit experiments. His methods typically employ perturbative expansions around an effective Hamiltonian where strong correlation effects are carefully incorporated through complex renormalizations. This provides greater flexibility but introduces complexities in accurately treating infinite correlation limits. Shastry, conversely, positions his ECFL formalism within a mathematically rigorous framework that explicitly encodes infinite correlation strength into the theory from its initial formulation. By invoking the Hubbard operator algebra, Shastry develops exact transformations, notably using exact Hubbard-Stratonovich transformations to convert complicated interaction terms into simpler, mathematically manageable forms.This strategy greatly facilitates the description of anomalous spectral functions and non-Fermi-liquid dynamics in a more controlled and theoretically rigorous manner.

Chapter 2

Strong Local Bosonic Fluctuations: The key to understanding strongly correlated metals

The work by Hassan, Prakash, Vidhyadhiraja, and Ramakrishnan [2] provides a detailed theoretical analysis aimed at understanding strongly correlated electron systems (SCES), particularly within the Extremely Correlated Fermi Liquid (ECFL) paradigm characterized by infinitely strong electron-electron interactions ($U = \infty$) described via the Hubbard model. Their approach uniquely emphasizes local bosonic fluctuations—charge and spin—as crucial drivers of electron dynamics, a departure from standard quasiparticle descriptions typical of conventional metals.

They start with the $U = \infty$ Hubbard Hamiltonian written in the X operator representation:

$$H = -\mu \sum_{i,\sigma} X_i^{\sigma\sigma} + \sum_{ij} t_{ij} X_i^{\sigma 0} X_j^{0\sigma} \tag{2.1}$$

with the aim to calculate the double-time retarded Green's function defined as:

$$G_{ij}^{R\sigma\sigma'}(t,t') = -i\theta(t-t') \left\langle \left\{ X_i^{0\sigma}(t), X_j^{\sigma' 0}(t') \right\}_+ \right\rangle \equiv \left\langle \left\langle X_i^{0\sigma}(t) \mid X_j^{\sigma' 0}(t') \right\rangle \right\rangle \tag{2.2}$$

and employ an equation-of-motion formalism to systematically derive self-consistent equations for the single-particle Green's function $G(\omega)$ and its associated self-energy $\Sigma(\omega)$. Crucially, their derivation explicitly involves bosonic correlation functions—local charge fluctuations (D_N) and spin fluctuations (D_S)—integrating them self-consistently into the Dyson equation:

$$G(\omega) = G_{\text{MF}}(\omega) + G_{\text{MF}}(\omega)\Sigma(\omega)G(\omega), \tag{2.3}$$

where $G_{\text{MF}}(\omega)$ denotes a mean-field Green's function and $\Sigma(\omega)$ is expressed through bosonic correlation functions involving both charge and spin dynamics. This framework introduces a nontrivial, numerically solvable self-consistency loop where the Green's function, self-energy, and bosonic correlations are iteratively computed until convergence is achieved.

Several approximations were involved in developing the self consistent equations like the $d = \infty$ approximation of self energy followed by a Non-crossing approximation(NCA) and Bose-Fermi or DG decoupling. An additional $d = \infty$ approximation in the current-current correlation function is made to drop the vertex functions. Under these considerations, the equation for self energy

$$\Sigma_{ij}^{R,\sigma\sigma}(\omega) = \frac{1}{Q^2} \sum_{ll'\sigma''\sigma'''} t_{jl} t_{il'} \left\langle \left\langle \delta B_i^{\sigma\sigma''}(t) X_{l'}^{0\sigma''}(t) \mid \delta B_j^{\dagger\,\sigma\sigma'''}(t') X_l^{\sigma'''0}(t') \right\rangle \right\rangle_{\omega}^{(p)} \tag{2.4}$$

$$= \frac{1}{Q^2} \sum_{ll'\sigma''\sigma'''} t_{jl} t_{il'} \left\langle \left\langle B_i^{\sigma\sigma''}(t) X_{l'}^{0\sigma''}(t) \mid B_j^{\dagger\,\sigma\sigma'''}(t') X_l^{\sigma'''0}(t') \right\rangle \right\rangle_{\omega} \tag{2.5}$$

reduces to

$$\Sigma_{ii}^{R,\sigma\sigma}(\omega) = \frac{1}{Q^2} \Bigg[-i\theta(t-t') \Bigg(\sum_{\sigma''} \left\langle B_i^{\sigma\sigma''}(t) B_i^{\dagger\,\sigma\sigma''}(t') \right\rangle \left\langle X_i^{0\sigma''}(t) X_i^{\sigma''0}(t') \right\rangle + \left\langle B_i^{\dagger\,\sigma\sigma''}(t') B_i^{\sigma\sigma''}(t) \right\rangle \left\langle X_i^{\sigma''0}(t') X_i^{0\sigma''}(t) \right\rangle \Bigg) \Bigg]_{\omega} \tag{2.6}$$

And using the spectral representation for the correlation function above, the self-energy expression is given by a convolution integral involving the spectral densities of the Green's function $\rho_G(\epsilon)$ and bosonic correlators $\rho_D(\epsilon)$:

$$\Sigma^R(\omega) = -\frac{1}{Q^2} \int d\epsilon_1 d\epsilon_2 \frac{\rho_G(\epsilon_1)\rho_D(\epsilon_2) \left[\tanh\left(\frac{\beta\epsilon_1}{2}\right) + \coth\left(\frac{\beta\epsilon_2}{2}\right)\right]}{\omega^+ - \epsilon_1 - \epsilon_2}. \tag{2.7}$$

A detailed self consistency loop and the Summary of equations involved is given below in the figure **??**

Their main findings highlight a crossover from a coherent Fermi liquid regime characterized by conventional quasiparticle excitations at low temperatures, into an incoherent quantum regime and eventually into a classical incoherent regime at higher temperatures as seen in 5.8a . The spectral function analysis shows a clear evolution from sharply defined quasiparticle peaks to broad, incoherent spectral weight distributions as tempera-

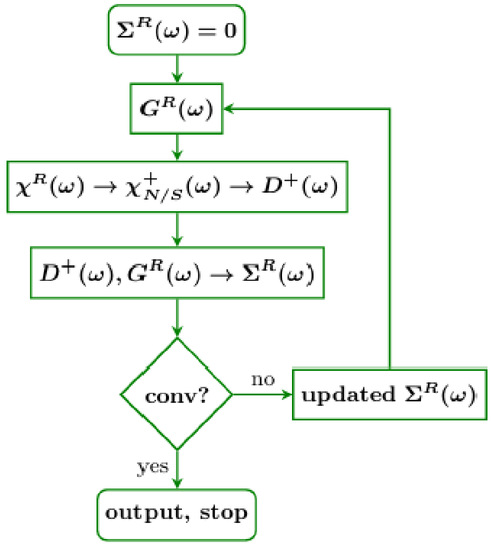


Figure 2.1: Self-Consistency Loop developed in [2]

$$G = G^{\mathrm{MF}} + G^{\mathrm{MF}}\Sigma G, \quad G^{\mathrm{MF}}(\omega) = \frac{Q}{\omega - Q\epsilon_k + i0^+}$$

$$\chi^R(\omega) = \frac{1}{N}\sum_k \iint d\omega_1 d\omega_2 \frac{\rho_G(k,\omega_1)\rho_G(k,\omega_2)v_k^2}{\omega + \omega_1 - \omega_2 + i\eta}[n_F(\omega_1) - n_F(\omega_2)]$$

$$\chi^R_N(\omega) = 2\chi^R(\omega) \quad \text{and} \quad \chi^R_S(\omega) = \chi^R(\omega)$$

$$\chi^\alpha_\gamma(\omega) = \int_{-\infty}^{\infty} d\omega' \frac{\left\{\frac{1+\alpha}{2} + n_B(\omega')\right\}\rho_\gamma(\omega')}{\omega - \omega' + i0^+}$$

$$\left[D^\alpha_N(\omega)\right]^{-1} = \alpha\frac{1}{n}\left(\omega - \alpha\frac{\chi^\alpha_N(\omega)}{n}\right), \quad \left[D^\alpha_S(\omega)\right]^{-1} = \alpha\frac{2}{n}\left(\omega - \alpha\frac{\chi^\alpha_S(\omega)}{\frac{n}{2}}\right)$$

$$D^R_\gamma(\omega) = \sum_\alpha D^\alpha_\gamma(\omega), \rho_{D_{N/S}}(\omega) = \sum_\alpha \rho^\alpha_{D_{N/S}}(\omega)$$

$$\rho_{D_{N/S}}(\omega) = -\rho_{D_{N/S}}(-\omega), \rho^-_{D_{N/S}}(\omega) = -e^{-\beta\omega}\rho^+_{D_{N/S}}(\omega),$$

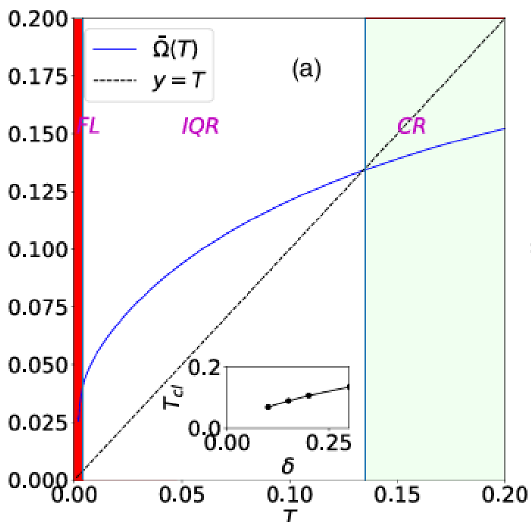
$$\Sigma^R(\omega) = -\frac{1}{Q^2}\int_{-\infty}^{\infty} d\epsilon_1 d\epsilon_2 \rho_G(\epsilon_1)\rho_D(\epsilon_2)\left(\frac{\tanh\left(\frac{\beta\epsilon_1}{2}\right) + \coth\left(\frac{\beta\epsilon_2}{2}\right)}{\omega^+ - \epsilon_1 - \epsilon_2}\right)$$

$$\text{where } \rho_G(\epsilon_1) = -\frac{1}{\pi}\mathrm{Im}G^R(\epsilon_1), \quad \rho_D(\epsilon_2) = -\frac{1}{\pi}\mathrm{Im}D^R(\epsilon_2).$$

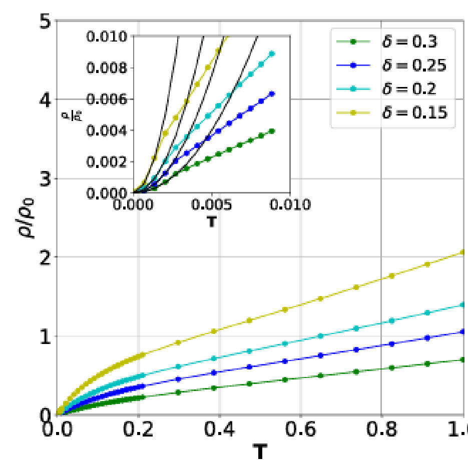
Figure 2.2: Summary of Equations in Self Consistency Loop

ture increases. Furthermore, the intrinsic dc resistivity $\rho(T)$ calculated within this framework exhibits the hallmark T^2-dependence at very low temperatures (consistent with Fermi liquid theory), transitioning to linear temperature dependence at higher temperatures—a prominent "strange metal" characteristic frequently observed experimentally in cuprates and other SCES5.8b.

One notable novelty in Hassan et al.'s approach is the explicit role assigned to local bosonic charge and spin fluctuations. They highlight the presence of a quantum diffusive mode, with its spectral function having a distinctly Lorentzian-like low-frequency form, representing damped, overdamped excitations without a restorative force. Mathematically, the damping factor $\Gamma(T)$, extracted from the charge correlation spectrum, plays a pivotal role and is found numerically to be roughly linear with temperature, thereby indicating a fundamental link between bosonic fluctuations and anomalous transport behaviors.



(a) Average frequency varies with temperature for a doping level of δ= 0.3



(a)

(b) Resistivity varies as T^2 at low temperatures transitioning to T dependence at higher temperature

Figure 2.3: Spin and Charge Diffusion and fluctuations and their role in Resistivity[2]

In contrast to B.S. Shastry's ECFL theory, which rigorously incorporates infinite correlation through a projection onto non-doubly occupied states using Hubbard operators in a fundamentally nonperturbative manner, the present approach by Hassan et al. shares some conceptual ground but differs methodologically by leveraging explicitly bosonic correlation functions derived through the equation-of-motion approach. Moreover, unlike Shastry's analytical exact-limit formulation, the current work relies heavily on numerically solvable iterative schemes, providing direct connections with experimentally measurable quantities such as spectral densities and resistivity.

Thus, Hassan et al. introduce a robust self-consistent formalism capable of describing complex behaviors in strongly correlated materials by highlighting the central role of local bosonic fluctuations. This approach enriches theoretical understanding and offers new avenues for exploring the universal and material-specific aspects of strongly correlated electron physics. While the self-consistency loop provide valuable insights into the local properties of strongly correlated systems, they are limited by their inability to fully capture non-local effects and complex collective behaviors. This is because the self consistency loop after all the approximations is capable of only giving local Greens Function and local self energy.

In this work, unconventional results emerge due to the interplay of charge and spin fluctuations, which lead to multiple temperature regimes. These regimes include both incoherent and classical behaviors, which are a result of the strong local correlations in the system.

Chapter 3

Ginzburg Landau theories for superconductivity

3.1 Conventional Ginzburg Landau theories and their applicability to High Tc Superconductors

Ginzburg-Landau (GL) theory, initially proposed by Vitaly Ginzburg and Lev Landau in 1950, provides a framework to describe superconductivity near the superconducting transition temperature T_c. The theory was purely phenomenological, based on symmetry considerations and experimental observations, and provided a framework to describe the macroscopic properties of superconductors without addressing the underlying microscopic mechanisms. The fundamental assumption in GL theory is that the superconducting state can be characterized by a complex order parameter, $\psi(\mathbf{r})$, whose magnitude and phase vary slowly in space thereby making it possible to coarse grain(zoom out) the fields and the order parameter. The assumptions are meaningful for BCS superconductors where the coherence length ξ is much larger($\sim 1000nm$) compared to lattice parameter($\sim 1nm$) The theory captures essential superconducting properties of conventional superconductors, including the coherence length ξ, penetration depth λ, and the nature of the superconducting transition (second-order) through a free-energy functional expanded near T_c [**?**, **?**]:

$$F_s = F_n + \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{1}{2m^*}\left|\left(-i\hbar\nabla - \frac{2e}{c}\mathbf{A}\right)\psi\right|^2 + \frac{|\mathbf{B}|^2}{8\pi}, \tag{3.1}$$

where α and β are temperature-dependent coefficients, m^* is the effective mass, $\mathbf{A}$ is the vector potential, and $\mathbf{B} = \nabla \times \mathbf{A}$ is the magnetic field. It is essential to keep in mind that

Despite its tremendous success in describing conventional superconductors close to T_c, the application of GL theory to high-temperature (high-T_c) superconductors, such as the cuprates, encounters serious conceptual and practical difficulties. The fundamental reason lies in the strongly correlated nature of these systems, which invalidates the core assumptions underpinning standard GL theory. Specifically, the GL theory assumes a smooth variation of the order parameter and a well-defined single-particle coherence length, conditions difficult to justify in the presence of strong electron-electron correlations and significant local fluctuations inherent in high-T_c superconductors [23, 5] where the coherence length ξ is just 2-3 times larger than the lattice parameter(~ 0.5nm)

Early attempts to adapt GL theory to HTSCs involved incorporating unconventional pairing symmetries, most prominently d-wave symmetry. Notable among these attempts is the work by Annett et al. [24], who derived a GL functional incorporating d-wave symmetry, intending to capture the pairing symmetry experimentally observed in cuprates. Similar phenomenological approaches were explored by Ren et al. [25], who explicitly constructed a GL-like theory that accounted for phase fluctuations, anisotropy, and the layered structure intrinsic to cuprate materials.

A comprehensive phenomenological theory was further proposed by Tesanovic and coworkers [26], who analyzed the extreme type-II superconductivity observed in cuprates near upper critical fields. Their GL-like model addressed critical fluctuations and vortex dynamics in strongly type-II superconductors, closely aligned with experimental data near H_{c2}. Additionally, the influential work by Franz and Millis [27] developed an approach based on GL-like formulations explicitly incorporating the role of phase fluctuations. Their studies successfully captured key features such as vortex liquid phases and significant deviations from mean-field predictions in underdoped cuprates.

Despite these sophisticated phenomenological extensions, microscopic derivations of GL theory from models like Hubbard or t-J models—more physically appropriate for strongly correlated HTSCs—faced profound challenges. Attempts to rigorously derive GL-type functionals from these strongly correlated models encountered serious difficulties due to nonlocal interactions and significant quantum fluctuations inherent in these systems. For instance, Gor'kov's traditional microscopic derivation of GL theory from the BCS Hamiltonian relies crucially on the weak-coupling and long coherence length approximations [28], conditions that fundamentally break down in cuprates. As highlighted by Emery and Kivelson [23], the extremely short coherence length and large phase fluctuations intrinsic to cuprates preclude any straightforward microscopic derivation akin to Gor'kov's methodology. Moreover, Varma's proposed marginal Fermi-liquid theory [29] and Sachdev's quantum-critical models [30], which attempt to reconcile GL-like phenomenological order parameters with quantum-critical fluctuations, illustrate further conceptual difficulties. Recently a mathematical study by R.L. Frank and M. Lemm [31] microscopically derived the GL Theory in the cases when multiple order paramters were involved. The aim of this study was to better understand the unconventional superconductors .

In essence, although numerous GL-like models have provided significant qualitative insights into high-T_c superconductivity, they inevitably fall short of achieving the predictive accuracy and microscopic justification that characterize conventional GL theory. These limitations stem primarily from the inherently strong correlation effects, nonlocality of interactions, and quantum fluctuations in HTSCs. Hence, the search for a truly effective GL-type theory for HTSCs remains open, likely necessitating fundamentally novel theoretical frameworks.

3.2 Phenomenological Ginzburg Landau-like Theory for superconductivity in the cuprates

In 2011, Banerjee et al. [3] developed a phenomenological Ginzburg-Landau-like (GL-like) theory to address the unique physics of cuprate superconductors. Unlike conventional Ginzburg-Landau (GL) theory, which assumes a smoothly varying, continuous superconducting order parameter near the transition temperature T_c, this theory explicitly places the superconducting order parameter on the bonds connecting Cu atoms. It thus captures strong local electronic correlations and short coherence lengths intrinsic to cuprates. Mathematically, the authors represent the free-energy functional as:

$$\mathcal{F}(\{\Delta_m, \phi_m\}) = \mathcal{F}_0(\{\Delta_m\}) + \mathcal{F}_1(\{\Delta_m, \phi_m\}), \tag{3.2}$$

$$\mathcal{F}_0(\{\Delta_m\}) = \sum_m \left(A\Delta_m^2 + \frac{B}{2}\Delta_m^4 \right), \tag{3.3}$$

$$\mathcal{F}_1(\{\Delta_m, \phi_m\}) = C \sum_{\langle mn \rangle} \Delta_m \Delta_n \cos(\phi_m - \phi_n). \tag{3.4}$$

where Δ_m and ϕ_m denote the amplitude and phase of the pairing order parameter defined on bond-centered sites. The phenomenological parameters A, B, and C have no microscopic origin and are empirically determined, based on experimental data. They have the following form:

$$A(x,T) = A_0\left[T - T_0\left(1 - \frac{x}{x_c}\right)\right]e^{T/T_p},$$

$$B = B_0 T_0,$$

$$C(x) = xC_0T_0.$$

The use of simple, GL-like functional that incorporates the nearest neighbour coupling term, that incorporates the anistropic nature of the d-wave order parameter is essentially heuristic and is valid over a broad $(x - T)$ region of cuprate phase diagram unlike the conventional GL Theories which are restricted by smallness and slow spatial variation of order parameter.

Employing mean-field approximations, Monte Carlo simulations, and numerical minimizations, the authors successfully replicated key experimental observations. Their calculations reproduced the dome-shaped doping dependence of T_c, the linear doping dependence of the superfluid stiffness at low doping (consistent with the Uemura relation), and demonstrated doping-dependent changes in vortex structures—from Josephson vortices at low doping to conventional BCS-like vortices at higher doping. The superfluid density ρ_s, a crucial measurable quantity, was explicitly calculated as:

$$\rho_s = -\frac{C}{2N_b}\left\langle \sum_{m,\mu} \Delta_m \Delta_{m+\mu} \cos(\phi_m - \phi_{m+\mu}) \right\rangle - \frac{C^2}{2N_bT}\sum_{\mu}\left\langle \Big(\sum_{m} \Delta_m \Delta_{m+\mu}\cos(\phi_m - \phi_{m+\mu})\Big)^2 \right\rangle \quad (3.5)$$

providing close quantitative agreement with experiments. The calculation of ρ_s also serves as a way to indicate the superconducting transition temperature(T_c) since it is above T_c that ρ_s becomes zero.

For very small doping and and doping in the proximity of Quantum Critical Point x_c the fluctuations are very large and low energy mobile electron degrees of freedom need to be considered explicitly thereby the authors add a term to the functional that includes quantum phase fluctuation effects:

$$\hat{\mathcal{F}}_Q(\{\hat{q}_m\}) = \frac{1}{2}\sum_{mn} \hat{q}_m V_{mn} \hat{q}_n \quad (3.6)$$

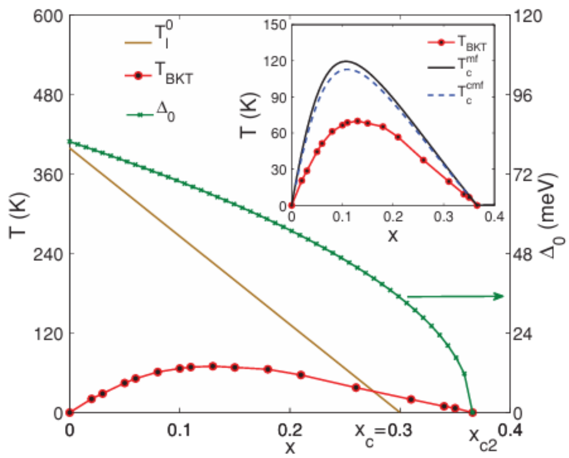


Figure 3.1: Phase Diagram generated from the Phenomenological theory[3]

here $\hat{q}_m$ is cooper pair number operator at site m and $\phi_m$is quantum mechanical operator conjugate to $\hat{q}_m$

A major success of Banerjee et al.'s theory is its natural prediction of two distinct temperature scales: a pseudogap crossover temperature T^*, where local pair amplitudes form, and the superconducting transition temperature T_c, at which global phase coherence emerges. Although their model does not explicitly invoke preformed Cooper pairs, it implicitly supports a similar picture, with local pairing amplitudes emerging significantly above T_c. Consequently, their framework accurately captures the widely observed pseudogap regime.

In summary, Banerjee et al.'s GL-like phenomenological theory significantly departs from conventional GL theory by incorporating lattice discreteness, strong local fluctuations, and clearly separated temperature scales for local pairing and global superconductivity. This extension was a significant development because it provided a more comprehensive understanding of HTS, explaining not only the superconducting state but also the anomalous behavior observed above the superconducting transition temperature. The ability to describe both the normal and superconducting states in a unified framework was a crucial advancement in understanding high-temperature superconductivity.

Considering these developments, it is meaningful to attempt a microscopic derivation of the GL-like free energy functional in the spirit of the extremely correlated fermi liquid regime and introducing the leading $(1/U)$ term as the perturbation to the Hamiltonian used by *Hassan et. al.*. The work although limited to calculating just localized Greens functions can in general also be used in the spirit of Dynamical Mean Field Theory(DMFT)[7]. The self-consistency loop can thereby be used to calculate

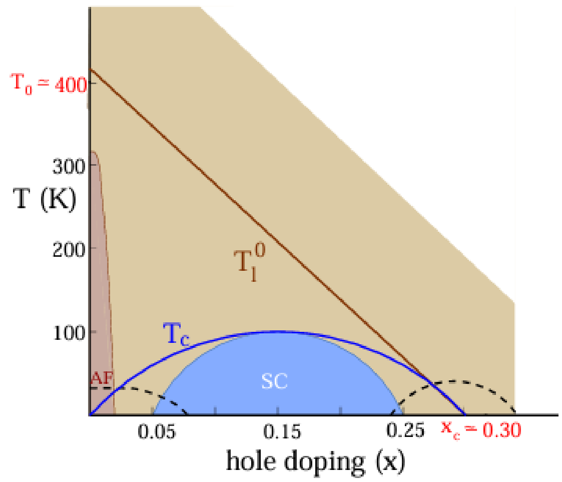


Figure 3.2: Phase Diagram of High Tc Superconductors

(a) Shaded region shows the region that was successfully explained by *Bannerjee et. al.*

the GL coefficients that have been microscopically derived in terms of Single particle Greens Functions.

Chapter 4

Microscopic Derivation of a Ginzburg Landau like Free Energy Functional

In this chapter I microscopically derive the free energy functional starting from the Hamiltonian and integrating the Fermionic Degrees of Freedom to eventually get a Free Energy Functional. This technique is commonly used in Field Theory approaches in condensed matter physics, for conventional superconductors, a similar derivation can be found in [32, 33] and is different from the conventional method of microscopically deriving the Ginzburg Landau theory using the Gor-kov Equations [28]. For d-wave superconductors as well similar attempts have been made[34].

I start by writing the 2D Hubbard Model in the Hubbard Operator basis followed by finding the effective low energy Hamiltonian using the Schrieffer-Wolf transformation to find the Low energy effective Hamiltoninan. This is followed by a Hubbard Stratanovich Tansformation and then we decide to go with the Cumulant Expansion of the resulting exponential term and considering processes upto fourth order, giving us a partition function in terms of the order parameter-(average spin-singlet nearest-neighbor Cooper pair amplitude) and the GL Coefficients written in terms of Operator expectation values in the $U = \infty$ Hamiltonian basis. We approximate the auxillary field to be site-dependent and time-independent which would be the appropriate setup to get the final equation of the desired form as mentioned in [3]. Reaching to the stage where the operator expectation value requires a decoupling approximation in the paramagnetic limit. This is necessary to write the expectation value in terms of single particle retarded double-time Greens Functions.

4.1 Effective Low Energy Hamiltonian

The Hamiltonian of the Hubbard model in the Hubbard operator representation when double occupancy is also included and chemical potential are also considered is given by:

$$H = \sum_{\langle i,j\rangle,\sigma} t_{ij}\left(X_i^{\sigma 0}X_j^{0\sigma}\right) - \mu\sum_{i,\sigma} X_i^{\sigma\sigma} + (U-2\mu)\sum_i X_i^{22} + \sum_{\langle i,j\rangle,\sigma} t_{ij}^1\left(X_i^{2\sigma}X_j^{\sigma 2}\right) + \sum_{\langle i,j\rangle,\sigma} t_{ij}^2\left(X_i^{2\bar{\sigma}}X_j^{0\sigma}\right) + \sum_{\langle i,j\rangle,\sigma} t_{ij}^3\left(X_i^{\sigma 0}X_j^{\bar{\sigma}2}\right) \quad (4.1)$$

where:

- t_{ij} is the hopping amplitude.
- σ represents the electron spin ($\sigma =\uparrow,\downarrow$).
- $X_i^{\sigma 0}$ is the Hubbard operator that annihilates an electron with spin σ at site i.
- $X_j^{0\sigma}$ is the Hubbard operator that creates an electron with spin σ at site j.
- U is the on-site Coulomb interaction.
- $X_i^{\sigma\sigma}$ is the projection operator for the doubly occupied state at site i.
- X_i^{22} is the projection operator for the doubly occupied state at site i.

Using the ideas discussed in [35] we split the Hamiltonian as follows:

$$H_0 = -\mu\sum_{i,\sigma} X_i^{\sigma\sigma} + \underbrace{\sum_{\langle i,j\rangle,\sigma} t_{ij}\left(X_i^{\sigma 0}X_j^{0\sigma}\right)}_{T_{00}} \quad (4.2)$$

$$H_{22} = (U-2\mu)\sum_i X_i^{22} \quad (4.3)$$

$$H_1 = \underbrace{\sum_{\langle i,j\rangle,\sigma} t_{ij}^1\left(X_i^{2\sigma}X_j^{\sigma 2}\right)}_{T_0} + \underbrace{\sum_{\langle i,j\rangle,\sigma} t_{ij}^2\left(X_i^{2\bar{\sigma}}X_j^{0\sigma}\right)}_{T_1} + \underbrace{\sum_{\langle i,j\rangle,\sigma} t_{ij}^3\left(X_i^{\sigma 0}X_j^{\bar{\sigma}2}\right)}_{T_{-1}} \quad (4.4)$$

In general the the nearest neighbor hopping amplitudes are not equal so we take three different Hopping amplitudes. Note that t^1, t^2, t^3 are just superscripts to distinguish the hopping amplitudes and not the powers of t.

These labels(underscripts) in H_1 correspond to whether the hopping term increases, decreases or keeps the no. of Double Occupancy sites constant. Now in order to get a Effective Low energy Hamiltonian we need to remove the terms T_1 and T_2 because they connect the Low and High Energy Subspace. We don't need to worry about T_0 term since it corresponds to a process keeping the system in High Energy Subspace only. The possibility of this process happening is very low in the $(U/t) >> 1$ limit and when projecting to the Low Energy Subspace the term is removed.

We use Schrieffer Wolf transformation, for this we first need to find the commutation relations for the different terms over here in the X operator representation. These commutation relations will help us to generate the

Following are the commutation relations derived in the X Operator formalism:

$$[\bar{H}_0, T_0] = 0 \tag{4.5}$$

$$[\bar{H}_0, T_1] = UT_1 \tag{4.6}$$

$$[\bar{H}_0, T_{-1}] = UT_{-1} \tag{4.7}$$

$$[\bar{H}_0, T_{00}] = 0 \tag{4.8}$$

Now these commutation relations are same as mentioned in [35] then we can use the same generator for Schrieffer Wolf transformation as was used by them.

So we take the Transformation Generator as follows:

$$S^{(2)} = U^{-1}(T_1 - T{-}1) + U^{-2}([T_1, T_0] + [T_{-1}, T_0]) \tag{4.9}$$

After using this Generator we have H_{eff} as follows (For this we can follow the exact same derivation that has been done in MGY to get H_{eff}, this was also done and verified):

$$H_{eff} = H_{00} + H_{22} + T_{00} + T_0 + U^{-1}([T_1 - T_{-1}]) + \mathcal{O}(t^3/U^2) \tag{4.10}$$

Now we have disconnected the Low Energy Subspace from the High Energy Subspace in $\mathcal{O}(t^3/U^2)$ order and now we move on to projecting to the Low Energy Subspace. This give the effective Low Energy Hamiltonian That has the form:

$$H_{eff}^{low} = H_{00} + T_{00} + U^{-1}([T_1, T_{-1}]) \tag{4.11}$$

Now the low energy subspace that we work with involves no occupancy and single occupancy states only. Considering this we expand the term $[T_1, T_{-1}]$:

$$[T_1, T_{-1}] = [\sum_{\langle i,j\rangle,\sigma} t_{ij}^2 \left(X_i^{2\bar{\sigma}} X_j^{0\sigma}\right), \sum_{\langle x,y\rangle,\sigma'} t_{xy}^3 \left(X_i^{\sigma' 0} X_j^{\overline{\sigma'}2}\right)] \tag{4.12}$$

$$[T_1, T_{-1}] = \sum_{\langle i,j\rangle,\sigma} \sum_{\langle x,y\rangle,\sigma'} [t_{ij}^2 X_i^{2\bar{\sigma}} X_j^{0\sigma}, t_{xy}^3 X_i^{\sigma' 0} X_j^{\overline{\sigma'}2}] \tag{4.13}$$

Just considering a single term inside the commutation for fixed i, j, x, y we get:

$$[t_{ij}^2 X_i^{2\bar{\sigma}} X_j^{0\sigma}, t_{xy}^3 X_x^{\sigma' 0} X_y^{\overline{\sigma'}2}] = t_{ij}^2 t_{xy}^3 (X_i^{2\bar{\sigma}} X_j^{0\sigma} X_x^{\sigma' 0} X_y^{\overline{\sigma'}2} - X_x^{\sigma' 0} X_y^{\overline{\sigma'}2} X_i^{2\bar{\sigma}} X_j^{0\sigma}) \tag{4.14}$$

Now the first term becomes zero because it involves double occupancy states as the initial and the final states (ideally speaking it should have been removed when we were projecting the system to Low energy subspace but this way). Now for the second term in this commutation relation, in case $y = i$ we stay in Low energy subspace and the double occupancy state is just a virtual intermediate state and thereby only in this case the second term survives otherwise it has the same fate as the first term- It would have to be removed ideally when we were projecting to the Low Energy Subspace. So the surviving term is :

$$[t_{ij}^2 X_i^{2\bar{\sigma}} X_j^{0\sigma}, t_{xy}^3 X_x^{\sigma' 0} X_y^{\overline{\sigma'}2}] = -\delta_{yi} t_{ij}^2 t_{xy}^3 (X_x^{\sigma' 0} X_y^{\overline{\sigma'}2} X_i^{2\bar{\sigma}} X_j^{0\sigma}) \tag{4.15}$$

$$\delta_{yi} t_{ij}^2 t_{xy}^3 (X_x^{\sigma' 0} X_y^{\overline{\sigma'}2} X_i^{2\bar{\sigma}} X_j^{0\sigma}) = -t_{ij}^2 t_{xi}^3 (X_x^{\sigma' 0} X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{0\sigma}) \tag{4.16}$$

Then the term$[T_1, T_{-1}]$:

$$[T_1, T_{-1}] = - \sum_{\langle i,j,x\rangle,\sigma\sigma'} t_{ij}^2 t_{xi}^3 (X_x^{\sigma' 0} X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{0\sigma}) \tag{4.17}$$

Now there are 2 possibilities :

1. If $x = j$ it becomes the conventional J term (Spin Exchange Term)of the T-J Model

2. If $x \neq j$ it is a 3 site term with the sites being next to each other(neighbouring sites), these 3 sites in a 2D Square lattice can form diagonal configuration resulting in interesting phenomenon

$$[T_1, T_{-1}] = -\sum_{\langle i,j\rangle,\sigma\sigma'} t_{ij}^2 t_{ji}^3 (X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{\sigma'\sigma}) - \underbrace{\sum_{\langle i,j\neq x\rangle,\sigma\sigma'} t_{ij}^2 t_{xi}^3 (X_x^{\sigma'0} X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{0\sigma})}_{\textbf{additional 3 site term}} \tag{4.18}$$

Then the Effective Low Energy Hamiltonian is:

$$H_{eff}^{low} = -\mu \sum_{i,\sigma} X_i^{\sigma\sigma} + \sum_{\langle i,j\rangle,\sigma} t_{ij} \left(X_i^{\sigma 0} X_j^{0\sigma} \right) - U^{-1}\Big(\sum_{\langle i,j\rangle,\sigma\sigma'} t_{ij}^2 t_{ji}^3 (X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{\sigma'\sigma}) + \sum_{\langle i,j\neq x\rangle,\sigma\sigma'} t_{ij}^2 t_{xi}^3 (X_x^{\sigma'0} X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{0\sigma})\Big) \tag{4.19}$$

Also for Hamiltonian to be Hermitian Conjugate $t_{ij}^{2*} = t_{ji}^3$

Then

$$H_{eff}^{low} = -\mu \sum_{i,\sigma} X_i^{\sigma\sigma} + \sum_{\langle i,j\rangle,\sigma} t_{ij} \left(X_i^{\sigma 0} X_j^{0\sigma} \right) - U^{-1}\Big(\sum_{\langle i,j\rangle,\sigma\sigma'} |t_{ij}^2|^2 (X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{\sigma'\sigma}) + \sum_{\langle i,j\neq x\rangle,\sigma\sigma'} t_{ij}^2 t_{xi}^3 (X_x^{\sigma'0} X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{0\sigma})\Big) \tag{4.20}$$

This 3 site term has been extensively discussed previously [36, 37]and from a numerical aspects in [11]. In X operator Formalism this term has been derived previously using a slightly different perturbative approach in the book by *Ovchinnikov and Val'Kov*[38](Chapter 5). They discuss this model in detail including the corresponding Feynman Diagrams and the contributioons to Self Energy of this term and call it the $T - J*$ Model.

If δ is the Hole density of the system then the conventional spin Exchange term is of order$(1 - \delta)t^2/U$ and 3 site term is of order $\delta t^2/U$. μ will also change according to the value of delta and will have to be determined Self consistently[39].

For now we ignore the 3 site term while carrying out the calculations further. Since it complicates the calculations and how to treat the 3 site term in the Functional integral theory is not very clear. Moreover we believe that the processes it induces can be taken care of

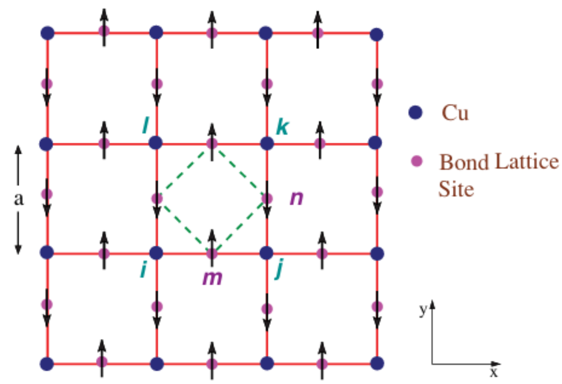


Figure 4.1: Bond Lattice Representation[3]

For the $t - J$ Model Hamiltonain in the X operator Basis is:

$$H^{low}_{eff} = -\mu \sum_{i,\sigma} X_i^{\sigma\sigma} + \sum_{\langle i,j \rangle,\sigma} t_{ij} \left(X_i^{\sigma 0} X_j^{0\sigma} \right) - U^{-1} (\sum_{\langle i,j \rangle,\sigma\sigma'} |t_{ij}^2|^2 (X_j^{\sigma' 0} X_i^{\overline{\sigma'} 2} X_i^{2\bar{\sigma}} X_j^{0\sigma})) \tag{4.21}$$

Now for writing in the ($J = |t|^2/U$ term in Double Lattice Formulation, first we expand it in Spin summation :

$$\sum_{\sigma\sigma'} t_{ij}^2 t_{ji}^3 (X_j^{\sigma' 0} X_i^{\overline{\sigma'}\bar{\sigma}} X_j^{0\sigma}) = t_{ij}^2 t_{ji}^3 (X_j^{\uparrow 0} X_i^{\downarrow\uparrow} X_j^{0\downarrow} + X_j^{\uparrow 0} X_i^{\downarrow\downarrow} X_j^{0\uparrow} + X_j^{\downarrow 0} X_i^{\uparrow\uparrow} X_j^{0\downarrow} + X_j^{\downarrow 0} X_i^{\uparrow\downarrow} X_j^{0\uparrow}) \tag{4.22}$$

If I use the following substitutions:

$$B_{ij}^{\dagger} = X_i^{\uparrow 0} X_j^{\downarrow 0} - X_j^{\uparrow 0} X_i^{\downarrow 0} = X_i^{\uparrow 0} X_j^{\downarrow 0} + X_i^{\downarrow 0} X_j^{\uparrow 0} \tag{4.23}$$

$$(B_{ij}^{\dagger})^{\dagger} = B_{ij} = X_j^{0\downarrow} X_i^{0\uparrow} + X_j^{0\uparrow} X_i^{0\downarrow} \tag{4.24}$$

then moving to Bond lattice lattice representation ($B_{ij} = B_m$) just for the spin exchange term, the term becomes:

$$B_m^{\dagger} B_m = B_{ij}^{\dagger} B_{ij} = (X_i^{\uparrow 0} X_j^{\downarrow 0} + X_i^{\downarrow 0} X_j^{\uparrow 0})(X_j^{0\downarrow} X_i^{0\uparrow} + X_j^{0\uparrow} X_i^{0\downarrow}) \tag{4.25}$$

This representation is just the dual of the site lattice representation expanding this

$$B_m^{\dagger} B_m = X_i^{\uparrow 0} X_j^{\downarrow 0} X_j^{0\downarrow} X_i^{0\uparrow} + X_i^{\uparrow 0} X_j^{\downarrow 0} X_j^{0\uparrow} X_i^{0\downarrow} + X_i^{\downarrow 0} X_j^{\uparrow 0} X_j^{0\downarrow} X_i^{0\uparrow} + X_i^{\downarrow 0} X_j^{\uparrow 0} X_j^{0\uparrow} X_i^{0\downarrow} \tag{4.26}$$

$$B_m^{\dagger} B_m = X_i^{\uparrow 0} X_j^{\downarrow\downarrow} X_i^{0\uparrow} + X_i^{\uparrow 0} X_j^{\downarrow\uparrow} X_i^{0\downarrow} + X_i^{\downarrow 0} X_j^{\uparrow\downarrow} X_i^{0\uparrow} + X_i^{\downarrow 0} X_j^{\uparrow\uparrow} X_i^{0\downarrow} \tag{4.27}$$

This has the exact same form as the expansion in equation(4.22) and thereby the substitutions made above can be used to bring the system in the bond lattice representation. This can be thought of as unconventional minimal coarse graining as is appropriate for cuprates due to their short coherence length ξ, where 2 site lattice points are replaced by a single bond lattice point, similar to one in the phenomenological GL-like theory [3] and was done keeping the free energy functional of this theory in mind, so that the order parameter can also be in bond lattice representation.

Now the Hamiltonian in the bond lattice-site lattice hybrid representation is:

$$H_{eff}^{low} = \underbrace{-\mu \sum_{i,\sigma} X_i^{\sigma\sigma} + \sum_{\langle i,j\rangle,\sigma} t_{ij} \left(X_i^{\sigma 0} X_j^{0\sigma} \right)}_{H_0} \underbrace{- U^{-1} (\sum_{\langle m\rangle} |t_m^2|^2 B_m^\dagger B_m)}_{H_I} \tag{4.28}$$

Notice that the H_0 is the Hamiltonian used by *Hassan et. al.*[2] and H_I which is the spin interaction term can be treated in the spirit of quantum perturbation.

4.2 Hubbard Stratonovich Transformation

To understand the effect of the $J = |t|^2/U$ coupling term when approaching the System from the Extremely correlated Limit seems to be a question of great interest. A new approach to answering some of these questions is to rephrase the self attraction term as a term describing linear coupling of Bfield to a time dependent Ψ field[40]. This can be done **exactly** using Hubbard Stratonovich Transformation(HST). The detailed derivation of this is now done. The subtelities and intricacies of the transformation are discussed in [41] and some other relevant information on how to proceed (since Coherent state path integrals are not that straightforward) are taken from Feynman Time ordering equation[42]. Although some resources which deal with path integrals in X operator representation are [43, 44]. Also there are ways to generate coherent states in these restricted Hilbert space as discussed in[[45]] which may be used to carry out the path integrals. However we stick to the approach of cumulant expansion where the exclusive need to find the coherent states and carry out path integrals is avoided. The relevant detailed transformation was followed mainly from[[40]]. Our Hamiltonian right now is of the form:

$$H_{eff}^{low} = \underbrace{-\mu \sum_{i,\sigma} X_i^{\sigma\sigma} + \sum_{\langle i,j\rangle,\sigma} t_{ij} \left(X_i^{\sigma 0} X_j^{0\sigma} \right)}_{H_0} - \underbrace{U^{-1}(\sum_{\langle m\rangle} |t_m^2|^2 B_m^\dagger B_m)}_{H_I} \tag{4.29}$$

In order to apply discretized HST need to shift to functional integral representation and bring into picture the Partition Function.

$$Z = \sum_i e^{-\beta E_1} \tag{4.30}$$

Now we do a Trotter decomposition,

$$Z = Tr(e^{-\beta(H_0+H_I)}) = Tr(\prod_{i=1}^{L} e^{-\Delta\tau(H_{0i}+H_{Ii})}) \approx Tr(\prod_{i=1}^{L} e^{-\Delta\tau H_{0i}} e^{-\Delta\tau H_{Ii}}) \tag{4.31}$$

Because $[H_{0i} + H_{Ii}, H_{0i+1} + H_{Ii+1}] = 0$ and in the last approximation there is a correction of order $\mathcal{O}(\Delta\tau^2[H_0, H_I])$. Now if we take to limit $N \to \infty$ or $\Delta\tau \to 0$. we can write:

$$Z = Tr(e^{-\beta H_0}\{T_\tau exp(-\int_0^\beta d\tau H_{I\tau})\}) \tag{4.32}$$

$$e^{-\beta H} = e^{-\beta H_0} T_\tau \left[\lim_{N\to\infty} \exp\left(-\beta \sum_{i=1}^{N} \frac{H_I(\tau_i)}{N} \right) \right] \tag{4.33}$$

$$= e^{-\beta H_0} T_\tau \left[\lim_{N\to\infty} \prod_{i=1}^{N} \exp\left(-\frac{\beta H_I(\tau_i)}{N} \right) \right] \tag{4.34}$$

This is Feynman Time Ordering Equation followed fromfeynman1951operator

Now following the HST as discussed in fletcher1990functional:

$$= e^{-\beta H_0} T_\tau \left[\lim_{N\to\infty} \prod_{i=1}^{N} \exp\left(\frac{\beta U^{-1} \sum_m t_m^2 t_m^3 B_{mi}^\dagger B_{mi})}{N} \right) \right] \tag{4.35}$$

$$= e^{-\beta H_0} T_\tau \left[\lim_{N\to\infty} \prod_{i=1}^{N} \exp\left(\frac{\beta (U^{-1} t^2 t^3) \sum_m B_{mi}^\dagger B_{mi})}{N} \right) \right] \tag{4.36}$$

$$= e^{-\beta H_0} T_\tau \left[\lim_{N\to\infty} \prod_{i=1}^{N} \exp\left(\frac{\beta J \sum_m B_{mi}^\dagger B_{mi})}{N} \right) \right] \tag{4.37}$$

We have replaced J by $|t|^2/U$.

And $B^{\dagger}_{mi} = B^{\dagger}_{m}(\tau_i)$ then at some particular i:

$$\beta J B^{\dagger}_{m} B_m = [(\sqrt{\beta J} B^{\dagger}_{m} + \sqrt{\beta J} B_m)^2 - (\sqrt{\beta J} B^{\dagger}_{m} - \sqrt{\beta J} B_m)^2]/4 \tag{4.38}$$

$$\begin{aligned} e^{\beta J B^{\dagger}_{m} B_m/N} &= e^{(\sqrt{\beta J} B^{\dagger}_{m} + \sqrt{\beta J} B_m)^2/4N - (\sqrt{\beta J} B^{\dagger}_{m} - \sqrt{\beta J} B_m)^2/4N} \\ &= e^{(\sqrt{\beta J} B^{\dagger}_{m} + \sqrt{\beta J} B_m)^2/4N} \times e^{-(\sqrt{\beta J} B^{\dagger}_{m} - \sqrt{\beta J} B_m)^2/4N} \end{aligned} \tag{4.39}$$

Even in the last step over here there is a $-1/2[A, B]$ term on breaking the 2 exponential terms which gets vanished[46] when $N \to \infty$. Thereby we need not worry about it.

$$\begin{aligned} &exp((\sqrt{\beta J} B^{\dagger}_{m} + \sqrt{\beta J} B_m)^2/4N) = \\ &\qquad \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dx_{mi}\, exp(-x^2_{mi}) exp(-x_{mi} \sqrt{\beta J} [B^{\dagger}_{m} + B_m]/\sqrt{N}) \end{aligned} \tag{4.40}$$

Similarly,

$$\begin{aligned} &exp(-(\sqrt{\beta J} B^{\dagger}_{m} - \sqrt{\beta J} B_m)^2/4N) = \\ &\qquad \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dy_{mi}\, exp(-y^2_{mi}) exp(-i y_{mi} \sqrt{\beta J} [B^{\dagger}_{m} - B_m]/\sqrt{N}) \end{aligned} \tag{4.41}$$

The product of these gives

$$\begin{aligned} &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dx_{mi}\, exp(-x^2_{mi}) exp(-x_{mi} \sqrt{\beta J} [B^{\dagger}_{m} + B_m]/\sqrt{N}) \\ &\qquad \times \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dy_{mi}\, exp(-y^2_{mi}) exp(-i y_{mi} \sqrt{\beta J} [B^{\dagger}_{m} - B_m]/\sqrt{N}) \end{aligned} \tag{4.42}$$

$$\begin{aligned} &= \frac{1}{\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_{mi} dy_{mi} exp(-\underbrace{(x^2_{mi} + y^2_{mi})}_{|\Psi_{mi}|^2}) \\ &\qquad \exp\left(-\sqrt{\frac{\beta J}{N}} (B^{\dagger}_{m} \underbrace{(x_{mi} + i y_{mi})}_{\Psi_{mi}} + B_m \underbrace{(x_{mi} - i y_{mi})}_{\Psi^*_{mi}}) \right) \end{aligned} \tag{4.43}$$

At this point it is meaningful to scale the auxillary fields as follows:

$$x_{mi} \to \frac{x_{mi}}{(N/\beta)^{1/2}}$$

This step is essential to get $d\tau$ in the exponents on taking the continuous limit. Also this ensures that the partition function is dimensionless. Thereby this substitution is necessary. This is because $d\Psi_{mi}$ also scales accordingly making it dimensionless

The final equation at a particular time index i:

$$= \frac{1}{\pi}\int_{-\infty}^{\infty} d\Psi_{mi} \exp\bigl(-\beta|\Psi_{mi}|^2/N\bigr) \exp\Bigl(-\beta\sqrt{J}(B_{mi}^{\dagger}\Psi_i + B_{mi}\Psi_i^*)/N\Bigr) \tag{4.44}$$

Putting this back in the equation 4.37 carefully with the product over m in ψ_m:

$$\begin{aligned}\exp(-\beta H) = e^{-\beta H_0} T_\tau \Bigg[\lim_{N\to\infty}\prod_{i=1}^{N}\frac{1}{\pi}\int_{-\infty}^{\infty}\prod_m d\Psi_{mi}\exp\Bigg(-\frac{\beta}{N}\sum_m |\Psi_{mi}|^2\Bigg) \\ \times \exp\Bigg(-\frac{\beta\sqrt{J}\sum_m\Big(B_{mi}^{\dagger}\Psi_{mi} + B_{mi}\Psi_{mi}^*\Big)}{N}\Bigg)\Bigg].\end{aligned} \tag{4.45}$$

Now taking the limit $N \to \infty$ we get the final form, **how does the time index on the differential element affect things**:

$$\begin{aligned}exp(-\beta H) = e^{-\beta H_0}\frac{1}{\pi}\int_{-\infty}^{\infty}\prod_m d\Psi_m \exp\Bigg(\int_0^\beta d\tau\sum_m -|\Psi_m(\tau)|^2\Bigg) \\ \times T_\tau\Bigg[\exp\Bigg(-\int_0^\beta d\tau\sum_m\sqrt{J}[B_m^{\dagger}(\tau)\Psi_m(\tau) + B_m(\tau)\Psi_m^*(\tau)]\Bigg)\Bigg]\end{aligned} \tag{4.46}$$

Thereby the final form after Hubbard Stratonovich transformation is:

$$\begin{aligned}Z = \Bigg\langle e^{-\beta H_0}\frac{1}{\pi}\int_{-\infty}^{\infty} d\Psi \exp\Bigg(-\int_0^\beta d\tau\sum_m |\Psi_m(\tau)|^2\Bigg) \\ \times T_\tau\Bigg[\exp\Bigg(-\int_0^\beta d\tau\sum_m\sqrt{J}[B_m^{\dagger}(\tau)\Psi_m(\tau) - B_m(\tau)\Psi_m^*(\tau)]\Bigg)\Bigg]\Bigg\rangle\end{aligned} \tag{4.47}$$

4.3 Cumulant Expansion

Most general equation of Cumulant expansion(eqn(3.5) from [47]):

$$\left\langle \exp\left\{ \int_a^b \sum_j X_j(t)dt \right\} \right\rangle = \exp\left\{ \sum_{n=1}^{\infty} \frac{1}{n!} \int_a^b dt_1 .... \int_a^b dt_n \sum_{j_1} .... \sum_{j_n} \langle X_{j_1}(t_1) ..... X_{j_n}(t_n) \rangle_c \right\} \quad (4.48)$$

Here X is a stochastic variable

Now taking

$$X_m(\tau) = B_m^\dagger(\tau)\Psi_m(\tau) + B_m(\tau)\Psi_m^*(\tau) \quad (4.49)$$

This is justified by the fact that the terms are not independent and the entire expression can be treated as a stochastic variable.

We get

$$\left\langle \exp\left[\int_0^\beta \sum_m B_m^\dagger(\tau)\Psi_m(\tau) + B_m(\tau)\Psi_m^*(\tau)\, d\tau \right] \right\rangle = \exp\left[\sum_{n=1}^{\infty} \frac{1}{n!} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_n \sum_{m_1} \cdots \sum_{m_n} \Big\langle (B_{m_1}^\dagger(\tau_1)\Psi_{m_1}(\tau_1) + B_{m_1}(\tau_1)\Psi_{m_1}^*(\tau_1)) \ldots \times (B_{m_n}^\dagger(\tau_n)\Psi_{m_n}(\tau_n) + B_{m_n}(\tau_n)\Psi_{m_n}^*(\tau_n)) \Big\rangle_c \right]. \quad (4.50)$$

Since we are looking for a GL-like functional(falling under the category of a ϕ^4 theory) we stick to expanding till $n = 4$ only. Now definitely for $n = 1$ and $n = 3$ the expectation values would be zero, Since there would be unequal number of creation and annihilation bond lattice operators. Thereby the expectation values of such terms calculated in the basis and dynamics of the Hamiltonian H_0 is always zero i.e., it does not increase or decrease the net number of average spin-singlet nearest-neighbor Cooper pair amplitude at each site.

For $n = 2$,

$$\frac{1}{2!}\int_0^\beta d\tau_1 d\tau_2 \sum_{m,m'} \left\langle \left(B_m^+(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1)\right)\left(B_{m'}^+(\tau_2)\,\psi_{m'}(\tau_2) + B_{m'}(\tau_2)\,\psi_{m'}^*(\tau_2)\right)\right\rangle_c \tag{4.51}$$

$$\begin{aligned}
&\left\langle \left(B_m^\dagger(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1)\right)\left(B_{m'}^\dagger(\tau_2)\,\psi_{m'}(\tau_2) + B_{m'}(\tau_2)\,\psi_{m'}^*(\tau_2)\right)\right\rangle_c = \\
&\quad \left\langle \left(B_m^+(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1)\right)\left(B_{m'}^+(\tau_2)\,\psi_{m'}(\tau_2) + B_{m'}(\tau_2)\,\psi_{m'}^*(\tau_2)\right)\right\rangle - \\
&\qquad \left\langle \left(B_m^+(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1)\right)\right\rangle \left\langle \left(B_{m'}^+(\tau_2)\,\psi_{m'}(\tau_2) + B_{m'}(\tau_2)\,\psi_{m'}^*(\tau_2)\right)\right\rangle
\end{aligned} \tag{4.52}$$

Now since $\langle (B_m^+(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1))\rangle = 0$. The second order cumulant becomes:

$$\begin{aligned}
&\left\langle \left(B_m^+(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1)\right)\left(B_{m'}^+(\tau_2)\,\psi_{m'}(\tau_2) + B_{m'}(\tau_2)\,\psi_{m'}^*(\tau_2)\right)\right\rangle_c = \\
&\left\langle \left(B_m^+(\tau_1)\,\psi_m(\tau_1) + B_m(\tau_1)\,\psi_m^*(\tau_1)\right)\left(B_{m'}^+(\tau_2)\,\psi_{m'}(\tau_2) + B_{m'}(\tau_2)\,\psi_{m'}^*(\tau_2)\right)\right\rangle
\end{aligned} \tag{4.53}$$

Now the terms surviving in second order term would be:

$$= \left\langle B_m^\dagger(\tau_1)\,B_{m'}(\tau_2)\,\psi_m(\tau_1)\,\psi_{m'}^*(\tau_2)\right\rangle + \left\langle B_m(\tau_1)\,\psi_m^*(\tau_1)\,B_{m'}^+(\tau_2)\,\psi_{m'}(\tau_2)\right\rangle \tag{4.54}$$

Now for the fourth order $n = 4$ term[$O(\psi^4)$],

A general fourth order cumlant expansion :

$$\begin{aligned}
\langle X_j X_k X_l X_m\rangle_c = {}& \langle X_j X_k X_l X_m\rangle \\
& - \big\{\langle X_j\rangle\langle X_k X_l X_m\rangle + \langle X_k\rangle\langle X_l X_j X_m\rangle \\
& + \langle X_l\rangle\langle X_m X_j X_k\rangle + \langle X_m\rangle\langle X_j X_k X_l\rangle\big\} \\
& - \big\{\langle X_j X_l\rangle\langle X_k X_m\rangle + \langle X_j X_m\rangle\langle X_k X_l\rangle + \langle X_k X_m\rangle\langle X_j X_l\rangle\big\} \\
& + 2\big\{\langle X_j X_k\rangle\langle X_l\rangle\langle X_m\rangle + \langle X_j X_l\rangle\langle X_k\rangle\langle X_m\rangle + \langle X_j X_m\rangle\langle X_k\rangle\langle X_l\rangle \\
& + \langle X_k X_l\rangle\langle X_j\rangle\langle X_m\rangle + \langle X_k X_m\rangle\langle X_j\rangle\langle X_l\rangle + \langle X_l X_m\rangle\langle X_j\rangle\langle X_k\rangle\big\} \\
& - 6\langle X_j\rangle\langle X_k\rangle\langle X_l\rangle\langle X_m\rangle.
\end{aligned} \tag{4.55}$$

Using this expansion for fourth order leads to a very long expansion, in our case only the term in the first line and the third line of 4.55 survive as only they can have both creation and annihilation operators in product:

$$\begin{aligned}\langle X_j X_k X_l X_m \rangle_c &= \langle X_j X_k X_l X_m \rangle \\ &\quad - \left\{ \langle X_j X_l \rangle \langle X_k X_m \rangle + \langle X_j X_m \rangle \langle X_k X_l \rangle + \langle X_k X_m \rangle \langle X_j X_l \rangle \right\}\end{aligned} \tag{4.56}$$

$$\begin{aligned}
&\Big\langle \big(B^+_m(\tau_1)\psi_m(\tau_1) - B_m(\tau_1)\psi^*_m(\tau_1)\big)\big(B^+_{m'}(\tau_2)\psi_{m'}(\tau_2) - B_{m'}(\tau_2)\psi^*_{m'}(\tau_2)\big) \\
&\quad \times \big(B^+_{m''}(\tau_3)\psi_{m''}(\tau_3) - B_{m''}(\tau_3)\psi^*_{m''}(\tau_3)\big)\big(B^+_{m'''}(\tau_4)\psi_{m'''}(\tau_4) - B_{m'''}(\tau_4)\psi^*_{m'''}(\tau_4)\big)\Big\rangle_c \\
&= \Big\langle \big(B^+_m(\tau_1)\psi_m(\tau_1) - B_m(\tau_1)\psi^*_m(\tau_1)\big)\big(B^+_{m'}(\tau_2)\psi_{m'}(\tau_2) - B_{m'}(\tau_2)\psi^*_{m'}(\tau_2)\big) \\
&\quad \times \big(B^+_{m''}(\tau_3)\psi_{m''}(\tau_3) - B_{m''}(\tau_3)\psi^*_{m''}(\tau_3)\big)\big(B^+_{m'''}(\tau_4)\psi_{m'''}(\tau_4) - B_{m'''}(\tau_4)\psi^*_{m'''}(\tau_4)\big)\Big\rangle \\
&\quad - \Bigg[\Big\langle \big(B^+_m(\tau_1)\psi_m(\tau_1) - B_m(\tau_1)\psi^*_m(\tau_1)\big)\big(B^+_{m'}(\tau_2)\psi_{m'}(\tau_2) - B_{m'}(\tau_2)\psi^*_{m'}(\tau_2)\big)\Big\rangle \\
&\qquad \times \Big\langle \big(B^+_{m''}(\tau_3)\psi_{m''}(\tau_3) - B_{m''}(\tau_3)\psi^*_{m''}(\tau_3)\big)\big(B^+_{m'''}(\tau_4)\psi_{m'''}(\tau_4) - B_{m'''}(\tau_4)\psi^*_{m'''}(\tau_4)\big)\Big\rangle \\
&\quad + \Big\langle \big(B^+_m(\tau_1)\psi_m(\tau_1) - B_m(\tau_1)\psi^*_m(\tau_1)\big)\big(B^+_{m''}(\tau_3)\psi_{m''}(\tau_3) - B_{m''}(\tau_3)\psi^*_{m''}(\tau_3)\big)\Big\rangle \\
&\qquad \times \Big\langle \big(B^+_{m'}(\tau_2)\psi_{m'}(\tau_2) - B_{m'}(\tau_2)\psi^*_{m'}(\tau_2)\big)\big(B^+_{m'''}(\tau_4)\psi_{m'''}(\tau_4) - B_{m'''}(\tau_4)\psi^*_{m'''}(\tau_4)\big)\Big\rangle \\
&\quad + \Big\langle \big(B^+_m(\tau_1)\psi_m(\tau_1) - B_m(\tau_1)\psi^*_m(\tau_1)\big)\big(B^+_{m'''}(\tau_4)\psi_{m'''}(\tau_4) - B_{m'''}(\tau_4)\psi^*_{m'''}(\tau_4)\big)\Big\rangle \\
&\qquad \times \Big\langle \big(B^+_{m'}(\tau_2)\psi_{m'}(\tau_2) - B_{m'}(\tau_2)\psi^*_{m'}(\tau_2)\big)\big(B^+_{m''}(\tau_3)\psi_{m''}(\tau_3) - B_{m''}(\tau_3)\psi^*_{m''}(\tau_3)\big)\Big\rangle\Bigg]
\end{aligned} \tag{4.57}$$

So the final equation for Free Energy Functional can be found by comparing

For a particular configuration of order parameter,

$$\begin{aligned}Z = \Bigg\langle e^{-\beta H_0} \frac{1}{\pi}\int_{-\infty}^{\infty} d\Psi_m \exp\Bigg(-\int_0^\beta d\tau \sum_m |\Psi(\tau)|^2\Bigg) \\ \times T_\tau\left[\exp\left(-\int_0^\beta d\tau \sum_m \sqrt{J}[B^\dagger_m(\tau)\Psi_m(\tau) + B_m(\tau)\Psi^*_m(\tau)]\right)\right]\Bigg\rangle\end{aligned} \tag{4.58}$$

For a particular configuration of order parameter,

$$Z(\{\psi_m\}) = \frac{1}{\pi}\Bigg\langle e^{-\beta H_0} e^{-\int_0^\beta d\tau \sum_m |\Psi_m(\tau)|^2} \\ \times T_\tau \left[\exp\left(-\int_0^\beta d\tau \sum_m \sqrt{J}[B_m^\dagger(\tau)\Psi_m(\tau) + B_m(\tau)\Psi_m^*(\tau)]\right)\right]\Bigg\rangle \quad (4.59)$$

Now, $< ... >_0 =< \rho_0 ... >$ where $\rho_0 = e^{-\beta H_0}/tr(e^{-\beta H_0})$ and $tr(e^{-\beta H_0}) = Z_0$ from [48]

$$Z(\{\psi_m\}) = \frac{Z_0}{\pi}\Bigg\langle e^{-\int_0^\beta d\tau \sum_m |\Psi_m(\tau)|^2} \\ \times T_\tau \left[\exp\left(-\int_0^\beta d\tau \sum_m \sqrt{J}[B_m^\dagger(\tau)\Psi_m(\tau) - B_m(\tau)\Psi_m^*(\tau)]\right)\right]\Bigg\rangle_0 \quad (4.60)$$

Now we can conveniently drop Z_0 and it is completely justified in context of GL Theory. It just adds a constant to free energy and has no role in defining dynamics. But it does shift absolute thermodynamic quantities that involve temperature derivatives of F itself. However if that is not the case, we can neglect it.

Now from Cyrot,1973[32] It is clear that $e^{\beta F(\{\psi_m\})} = Z(\{\psi_m\})$ when they say,

$$Z = \exp(-\beta F) = \int d\psi(r) \exp\left(-\beta \int F(\psi) d^3r\right)$$

Thererby we can write,

$$e^{-\beta F(\{\psi_m\})} = e^{-\int_0^\beta d\tau \sum_m |\Psi(\tau)|^2} \\ \times \exp\left[\frac{1}{2!}\int_0^\beta d\tau_1 d\tau_2 \sum_{m,m'} J\left\langle B_m^\dagger(\tau_1) B_{m'}(\tau_2) \psi_m(\tau_1) \psi_{m'}^*(\tau_2)\right\rangle_0 \right. \\ \left. + J\left\langle B_m(\tau_1) B_{m'}^+(\tau_2) \psi_m^*(\tau_1) \psi_{m'}(\tau_2)\right\rangle_0 + \frac{J^2}{4!} O(\psi^4)\right] \quad (4.61)$$

Here ψ_m is the complex valued order parameter (OP)- average spin-singlet nearest-neighbor Cooper pair amplitude given by

$$\langle B_m\rangle = \langle B_{ij}\rangle = \langle X_i^{\uparrow 0} X_j^{\downarrow 0} - X_i^{\downarrow 0} X_j^{\uparrow 0}\rangle$$

Three cases are possible for the order parameter:

- Site and time independent
- Site dependent and time independent
- Site and time dependent (Most General)

Now I assume the order parameter/auxillary field to be time independent but site dependent- again motivated by the Free Energy form of the phenomenological GL Theory [3] and a necessity for cuprate superconductors where the OP has d-wave symmetry.

4.3.1 Time independent Site dependent auxillary field

The complex auxillary field is of the form:

$$\psi_m = \Delta_m e^{i\phi_m}$$

Focusing on the second order terms for now,

The terms that would survive after taking the long wavelength limit/local limit (This is an asserted approximation) are (Write in appendix):

1) When $m' = m$

$$\sum_m \left\langle B_m^\dagger(\tau_1) B_m(\tau_2) \right\rangle_0 \Delta_m \Delta_m + \sum_m \left\langle B_m(\tau_1) B_m^\dagger(\tau_2) \right\rangle_0 \Delta_m \Delta_m \tag{4.62}$$

$$\begin{aligned}
&\left\langle B_m^\dagger(\tau_1) B_m(\tau_2) \right\rangle_0 \\
&= \left\langle \left(X_i^{\uparrow 0}(\tau_1) X_j^{\downarrow 0}(\tau_1) + X_i^{\downarrow 0}(\tau_1) X_j^{\uparrow 0}(\tau_1) \right) \left(X_j^{0\downarrow}(\tau_2) X_i^{0\uparrow}(\tau_2) + X_j^{0\uparrow}(\tau_2) X_i^{0\downarrow}(\tau_2) \right) \right\rangle_0 \\
&\qquad = \Big\langle X_i^{\uparrow 0}(\tau_1) X_j^{\downarrow 0}(\tau_1) X_j^{0\downarrow}(\tau_2) X_i^{0\uparrow}(\tau_2) + X_i^{\uparrow 0}(\tau_1) X_j^{\downarrow 0}(\tau_1) X_j^{0\uparrow}(\tau_2) X_i^{0\downarrow}(\tau_2) \\
&\qquad + X_i^{\downarrow 0}(\tau_1) X_j^{\uparrow 0}(\tau_1) X_j^{0\downarrow}(\tau_2) X_i^{0\uparrow}(\tau_2) + X_i^{\downarrow 0}(\tau_1) X_j^{\uparrow 0}(\tau_1) X_j^{0\uparrow}(\tau_2) X_i^{0\downarrow}(\tau_2) \Big\rangle_0
\end{aligned} \tag{4.63}$$

We **assert** the following decoupling approximation, it was necessary as we just have the equations of motion to calculate the single particle retarded greens functions. proper argument to justify this decoupling is still not very clear and this approximation is essentially heuristic in nature. A detailed discussion on the approximations is in the AppendixA

$$\left\langle X_i^{\uparrow 0}(\tau_1) X_i^{0\uparrow}(\tau_2)\right\rangle \left\langle X_j^{\downarrow 0}(\tau_1) X_j^{0\downarrow}(\tau_2)\right\rangle + \left\langle X_i^{\uparrow 0}(\tau_1) X_i^{0\downarrow}(\tau_2)\right\rangle \left\langle X_j^{\downarrow 0}(\tau_1) X_j^{0\uparrow}(\tau_2)\right\rangle + \left\langle X_i^{\downarrow 0}(\tau_1) X_i^{0\uparrow}(\tau_2)\right\rangle \left\langle X_j^{\uparrow 0}(\tau_1) X_j^{0\downarrow}(\tau_2)\right\rangle + \left\langle X_i^{\downarrow 0}(\tau_1) X_i^{0\downarrow}(\tau_2)\right\rangle \left\langle X_j^{\uparrow 0}(\tau_1) X_j^{0\uparrow}(\tau_2)\right\rangle \quad (4.64)$$

$$\begin{aligned} &\left\langle X_i^{\uparrow 0}(\tau_1) X_i^{0\uparrow}(\tau_2)\right\rangle \left\langle X_j^{\uparrow 0}(\tau_1) X_j^{0\uparrow}(\tau_2)\right\rangle + <><> \\ &\quad + <><> + \left\langle X_i^{\downarrow 0}(\tau_1) X_i^{0\downarrow}(\tau_2)\right\rangle \left\langle X_j^{\downarrow 0}(\tau_1) X_j^{0\downarrow}(\tau_2)\right\rangle \\ &\quad = G_{jj}^{\downarrow\downarrow}(\tau_1,\tau_2)\, G_{ii}^{1\uparrow}(\tau_1,\tau_2) + G_{jj}^{\uparrow\uparrow}(\tau_1,\tau_2)\, G_{ii}^{\downarrow\downarrow}(\tau_1,\tau_2) \end{aligned} \quad (4.65)$$

The terms mentioned as <><> correspond to some spin exchange process which gives Greens Functions that are simply zero in the paramagnetic spin isotropic regime in which the $U = \infty$ theory was developed[[2]].

Similarly,

$$\left\langle B_m(\tau_1) B_m^{\dagger}(\tau_2)\right\rangle_0 =$$

$$\begin{aligned} &\left\langle X_j^{0\downarrow}(\tau_1)\, X_j^{\downarrow 0}(\tau_2)\right\rangle \left\langle X_i^{0\uparrow}(\tau_1)\, X_i^{\uparrow 0}(\tau_2)\right\rangle + <><> \\ &\quad + <><> + \left\langle X_j^{0\uparrow}(\tau_1)\, X_j^{\wedge 0}(\tau_2)\right\rangle \left\langle X_i^{0\downarrow}(\tau_1)\, X_i^{\iota 0}(\tau_2)\right\rangle \\ &\quad = G_{jj}^{\downarrow\downarrow}(\tau_2,\tau_1)\, G_{ii}^{1\uparrow}(\tau_2,\tau_1) + G_{jj}^{\uparrow\uparrow}(\tau_2,\tau_1)\, G_{ii}^{\downarrow\downarrow}(\tau_2,\tau_1) \end{aligned} \quad (4.66)$$

Keep in mind that there is a time ordered operator acting on the entire term Eg. 4.47. Thereby these are Normal time Greens Functions.

These Greens functions are spin independent $G^{\sigma\sigma'} = G$. Now I assert that the Greens function is translation invariant which is essentially true since there are no disorders in the system and the doping is considered to be uniform throughout the lattice.

$$G_{ij}(\tau_1 - \tau_2) = G(i - j, \tau_1 - \tau_2)$$
$$G_{ij}(\tau_1 - \tau_2) = \frac{1}{V}\sum_k G_k(\tau_1 - \tau_2) e^{-ik\cdot(i-j)}$$

Applying this to the previous equations, it becomes :

$$\left\langle B_m^\dagger(\tau_1)B_m(\tau_2)\right\rangle_0 = 2G\left(0,\tau_1-\tau_2\right)G\left(0,\tau_1-\tau_2\right) \tag{4.67}$$

and

$$\left\langle B_m(\tau_1)B_m^\dagger(\tau_2)\right\rangle_0 = 2G\left(0,\tau_2-\tau_1\right)G\left(0,\tau_2-\tau_1\right) \tag{4.68}$$

Now this quantity can be taken outside the summation in 4.62 and this give the following quantities, with a change of time variable in any one of the terms. :

$$\int_0^\beta d\tau_1 d\tau_2 \Big[\sum_m \left\langle B_m^\dagger(\tau_1)B_m(\tau_2)\right\rangle_0 \Delta_m\Delta_m + \sum_m \left\langle B_m(\tau_1)B_m^\dagger(\tau_2)\right\rangle_0 \Delta_m\Delta_m\Big] = 4\int_0^\beta d\tau_1 d\tau_2 G\left(0,\tau_2-\tau_1\right)G\left(0,\tau_2-\tau_1\right)\sum_m |\Delta_m|^2 \tag{4.69}$$

Taking a fourier Transform to momentum space:

$$4\int_0^\beta d\tau_1 d\tau_2 \frac{1}{V^2}\sum_{k,k'} G_k\left(\tau_2-\tau_1\right)G_{k'}\left(\tau_2-\tau_1\right)\sum_m |\Delta_m|^2 \tag{4.70}$$

Taking a very similar approach for the nearest neighbour site terms

2) Nearest neighboring Site $m' = n$

$$\int_0^\beta d\tau_1 d\tau_2 \Big[\sum_{<m,n>} \left\langle B_m^\dagger(\tau_1)B_n(\tau_2)\right\rangle_0 \Delta_m\Delta_n e^{i(\phi_m-\phi_n)} + \sum_m \left\langle B_m(\tau_1)B_n^\dagger(\tau_2)\right\rangle_0 \Delta_m\Delta_n e^{-i(\phi_m-\phi_n)}\Big] \tag{4.71}$$

$$e^{-\beta\sum_m|\Delta_m|^2}\Big[\sum_{<m,n>}\Delta_m\Delta_n\int_0^\beta d\tau_1 d\tau_2 \{\left\langle B_m^\dagger(\tau_1)B_n(\tau_2)\right\rangle_0 e^{i(\phi_m-\phi_n)} + \sum_m \left\langle B_m(\tau_1)B_n^\dagger(\tau_2)\right\rangle_0 e^{-i(\phi_m-\phi_n)}\}\Big] \tag{4.72}$$

$$
\begin{aligned}
&\left\langle B_m^\dagger(\tau_1) B_n(\tau_2)\right\rangle_0 = \left\langle \left(X_i^{\uparrow 0} X_j^{\downarrow 0} + X_i^{\downarrow 0} X_j^{\uparrow 0}\right)(\tau_1) \left(X_k^{0\downarrow} X_j^{0\uparrow} + X_k^{0\uparrow} X_j^{0\downarrow}\right)(\tau_2)\right\rangle_0 \\
&= \left\langle X_i^{\uparrow 0} X_j^{\downarrow 0} X_k^{0\downarrow} X_j^{0\uparrow} + X_i^{\uparrow 0} X_j^{\downarrow 0} X_k^{0\uparrow} X_j^{0\downarrow} + X_i^{\downarrow 0} X_j^{\uparrow 0} X_k^{0\downarrow} X_j^{0\uparrow} + X_i^{\downarrow 0} X_j^{\uparrow 0} X_k^{0\uparrow} X_j^{0\downarrow}\right\rangle_0
\end{aligned}
\tag{4.73}
$$

Again using the same decoupling approximation:

$$
\begin{aligned}
&<><> + \left\langle X_i^{\uparrow 0}(\tau_1) X_k^{0\uparrow}(\tau_2)\right\rangle \left\langle X_j^{\uparrow 0}(\tau_1) X_j^{0\uparrow}(\tau_2)\right\rangle + \\
&\qquad \left\langle X_i^{\downarrow 0}(\tau_1) X_k^{0\downarrow}(\tau_2)\right\rangle \left\langle X_j^{\downarrow 0}(\tau_1) X_j^{0\downarrow}(\tau_2)\right\rangle + <><>
\end{aligned}
\tag{4.74}
$$

Each site m in dual lattice representation in 2 dimensions has 4 neighbors, call them $n1, n2, n3, n4$.

Following are the calculations for n1 and corresponding lattice sites (i_1, j_1)

$$
\begin{aligned}
&\left\langle B_m^\dagger(\tau_1) B_{n_1}(\tau_2)\right\rangle_0 = G_{ij_1}^{\uparrow\uparrow}(\tau_1,\tau_2)\, G_{ji_1}^{\downarrow\downarrow}(\tau_1,\tau_2) + G_{ij_1}^{\downarrow\downarrow}(\tau_1,\tau_2)\, G_{ji_1}^{\uparrow\uparrow}(\tau_1,\tau_2) \\
&= 2G(i-j_1, \tau_1-\tau_2) G(j-i_1, \tau_1-\tau_2) = 2G(0,\tau_1-\tau_2) G(-ax+ay, \tau_1-\tau_2)
\end{aligned}
\tag{4.75}
$$

and

$$
\begin{aligned}
&\left\langle B_m(\tau_1) B_{n_1}^\dagger(\tau_2)\right\rangle_0 = G_{i_1 j}^{\downarrow\downarrow}(\tau_2,\tau_1)\, G_{j_1 i}^{1\uparrow}(\tau_2,\tau_1) + G_{i_1 j}^{\uparrow\uparrow}(\tau_2,\tau_1)\, G_{j_1 i}^{\downarrow\downarrow}(\tau_2,\tau_1) \\
&= 2G(i_1-j, \tau_2-\tau_1) G(j_1-i, \tau_2-\tau_1) = 2G(ax-ay,\tau_2-\tau_1) G(0, \tau_2-\tau_1)
\end{aligned}
\tag{4.76}
$$

Then the equation 4.71 becomes

$$
\begin{aligned}
&= \int_0^\beta d\tau_1 d\tau_2 \Big[2G(0,\tau_1-\tau_2) G(-ax+ay,\tau_1-\tau_2) \sum_m \Delta_m \Delta_{n_1} e^{i(\phi_m-\phi_{n_1})} + \\
&\qquad 2G(ax-ay,\tau_2-\tau_1) G(0,\tau_2-\tau_1) \Delta_m \Delta_{n_1} e^{-i(\phi_m-\phi_{n_1})}
\end{aligned}
\tag{4.77}
$$

Now Fourier Tranform the Greens Function to momentum space,

$$
G(-ax+ay, \tau_1-\tau_2) = \frac{1}{V}\sum_k G_k(\tau_1-\tau_2) e^{-ik_x a + ik_y a}, G(0,\tau_1-\tau_2) = \frac{1}{V}\sum_{k'} G_{k'}(\tau_1-\tau_2)
\tag{4.78}
$$

Similar Fourier Transform for the other term as well

$$\int_0^\beta d\tau_1 d\tau_2 \Big[\frac{2}{V^2}\sum_{k,k'} G_k(\tau_1-\tau_2)e^{-ik_xa+ik_ya}G_{k'}(\tau_1-\tau_2)\sum_m \Delta_m\Delta_{n_1}e^{i(\phi_m-\phi_{n_1})}$$
$$+\frac{2}{V^2}\sum_{k,k'} G_k(\tau_2-\tau_1)e^{ik_xa-ik_ya}G_{k'}(\tau_2-\tau_1)\sum_m \Delta_m\Delta_{n_1}e^{-i(\phi_m-\phi_{n_1})} \quad (4.79)$$

Now since τ_1, τ_2 are dummy variables,can be changed in second term. Also in second term, I make the assertion that $G_k = G_{-k}$ which is completely justified for Greens Functions for 2D lattice in DMFT framework where the Self energy is local and momentum independent:

$$G_k(\omega) = \frac{1}{\omega - \epsilon_k + \mu - \sum(\omega)} \quad (4.80)$$

where $\epsilon_k = -2t(cos(k_x) + cos(k_y))$

Using this assertion, we get

$$\int_0^\beta d\tau_1 d\tau_2 \Big[\frac{4}{V^2}\sum_{k,k'} G_k(\tau_1-\tau_2)e^{-ik_xa+ik_ya}G_{k'}(\tau_1-\tau_2)\sum_m \Delta_m\Delta_{n_1}cos(\phi_m-\phi_{n_1})) \quad (4.81)$$

Now for the other neighbors $n2, n3, n4$. Using the form as in 4.80, we observe that

$$G_{k_x,k_y} = G_{k_x,-k_y} = G_{-k_x,k_y} = G_{-k_x,-k_y}$$

for some particular k_x, k_y. Then it is easy to see that the coefficient for all the 4 neighbours would be exactly same.

Thereby the entire neighbouring site term becomes,

$$\int_0^\beta d\tau_1 d\tau_2 \Big[\sum_{<m,n>} \left\langle B_m^\dagger(\tau_1)B_n(\tau_2)\right\rangle_0 \Delta_m\Delta_n e^{i(\phi_m-\phi_n)} +$$
$$\sum_{<m,n>} \left\langle B_m(\tau_1)B_n^\dagger(\tau_2)\right\rangle_0 \Delta_m\Delta_n e^{-i(\phi_m-\phi_n)}\Big]$$
$$= \Big(\int_0^\beta d\tau_1 d\tau_2 \frac{4}{V^2}\sum_{k,k'} G_k(\tau_1-\tau_2)e^{-ik_xa+ik_ya}G_{k'}(\tau_1-\tau_2)\Big)\sum_{<m,n>} \Delta_m\Delta_n cos(\phi_m-\phi_n)) \quad (4.82)$$

In conclusion, we take the long wavelength/local approximation approximation, we claim that only 2 terms will survive the onsite term and the nearest neighborhood hopping term ($m' = m$ and $m' = n$ where n is a neighbor). This approximation is essential to get the Free Energy Functional form of desired form[3]. Moreover such a approximation is necessary for the coarse graining relevant to Ginzburg Landau theories claiming slow variations at these lengths and the fact that onsite and the nearest neighbour term are enough to capture all the essential physics.

3) Simplest fourth order term,

In a conventional ϕ^4 theory, the fourth order term is not treated in extensive mathematical detail and a rigorous treatment is not done. Even for the BCS Theory, they consider just the simplest term, not even the first order corrections were considered [49]. Altland and Simons[50] say:

"For temperatures below T_c, the quadratic action becomes unstable and – in direct analogy with our previous discussion of the superfluid condensate action – we have to turn to the fourth-order contribution, $S^{(4)}$, to ensure stability of the functional integral. At orders $n > 2$ of the expansion, spatial and temporal gradients can be safely neglected (due to the smallness of $\Delta \ll T$, they will certainly be smaller than the gradient contributions to $S^{(2)}$)."

Thereby it is best to stick to the simplest term in fourth order for now ($m = m' = m'' = m'''$) with some dummy variable changes.

$$
\begin{aligned}
&\Big\langle \big(B_m^+(\tau_1)\psi_m + B_m(\tau_1)\psi_m^*\big)\big(B_m^+\psi_m + B_m(\tau_2)\psi_m^*\big) \\
&\quad \times \big(B_m^+(\tau_3)\psi_m + B_m(\tau_3)\psi_m^*\big)\big(B_m^\dagger(\tau_1)\psi_m + B_m(\tau_4)\psi_m^*\big)\Big\rangle_c \\
&= \Big\langle \big(B_m^+(\tau_1)\psi_m + B_m(\tau_1)\psi_m^*\big)\big(B_m^+\psi_m + B_m(\tau_2)\psi_m^*\big) \\
&\quad \times \big(B_m^+(\tau_3)\psi_m + B_m(\tau_3)\psi_m^*\big)\big(B_m^\dagger(\tau_1)\psi_m + B_m(\tau_4)\psi_m^*\big)\Big\rangle \\
&\quad - 3\Bigg[\Big\langle \big(B_m^+(\tau_1)\psi_m + B_m(\tau_1)\psi_m^*\big)\big(B_m^+(\tau_2)\psi_m + B_m(\tau_2)\psi_m^*\big)\Big\rangle \\
&\qquad \times \Big\langle \big(B_m^+(\tau_3)\psi_m + B_m(\tau_3)\psi_m^*\big)\big(B_m^+(\tau_4)\psi_m - B_m(\tau_4)\psi_m^*\big)\Bigg]
\end{aligned} \tag{4.83}
$$

Only the terms having equal no of creation and annihilation operators will survive . Over that I also decouple the fourth order terms :

$$\langle B_m^\dagger(\tau_1)B_m(\tau_2)B_m(\tau_3)B_m^\dagger(\tau_4)\rangle = \langle B_m^\dagger(\tau_1)B_m(\tau_2)\rangle\langle B_m^\dagger(\tau_4)B_m(\tau_3)\rangle \tag{4.84}$$

In total there would be 6 decoupling terms 4_{C_2}. On decoupling all will give the same quantity. Also there is a negative part of 4.83 which will also give the exact same quantity. Then using the same approach as used in the second order terms, the expression for simplest fourth order term is :

$$\frac{6}{V^4}\int_0^\beta d\tau_1 d\tau_2 d\tau_3 d\tau_4 \sum_{k,k',k'',k'''} G_k(\tau_1-\tau_2)G_{k'}(\tau_1-\tau_2)G_{k''}(\tau_3-\tau_4)G_{k'''}(\tau_3-\tau_4)\sum_m |\Delta_m|^4 \tag{4.85}$$

Putting the calculated forms of second and fourth order terms, the final free energy functional after carrying out a cumulant expansion is:

$$\begin{aligned} e^{-\beta F(\{\Delta_m,\phi_m\})} &= e^{-\beta\sum_m|\Delta_m|^2} \\ &\times \exp\left[\frac{1}{2!}\Big[4\int_0^\beta d\tau_1 d\tau_2 \frac{J}{V^2}\sum_{k,k'} G_k\left(\tau_2-\tau_1\right)G_{k'}\left(\tau_2-\tau_1\right)\sum_m|\Delta_m|^2\right. \\ &+\Big(\int_0^\beta d\tau_1 d\tau_2 \frac{4J}{V^2}\sum_{k,k'} G_k(\tau_1-\tau_2)e^{-ik_x a+ik_y a}G_{k'}(\tau_1-\tau_2)\Big)\sum_{<m,n>}\Delta_m\Delta_n cos(\phi_m-\phi_n))\Big] \\ &+\frac{1}{4!}\frac{6J^2}{V^4}\int_0^\beta d\tau_1 d\tau_2 d\tau_3 d\tau_4 \sum_{k,k',k'',k'''} G_k(\tau_1-\tau_2)G_{k'}(\tau_1-\tau_2)G_{k''}(\tau_3-\tau_4)G_{k'''}(\tau_3-\tau_4)\sum_m|\Delta_m|^4\Big] \end{aligned} \tag{4.86}$$

Observe the Free Energy Functional over here has a positive coefficient for the fourth order term. This just makes the partition function divergent and unnatural thereby it is meaningful to multiply the entire quantity with a -ve sign and it also makes perfect sense with the phenomenology in that case. This is a general problem that shows up in microscopoic derivations of Free Energy Functional when we couple the auxillary fields with fermionic degrees of freedom under some special conditions and is termed Parity Anomaly[51, 52]. More details about the mathematical treatment and its validity in the present context can be found in the Appendix B

So I replace $F_{micro} \to -F_{micro}$. Now it converges and a physically meaningful partition function is achieved. Then the Free Energy Functional is :

$$F(\{\Delta_m,\phi_m\}) = \Big[\Big(\frac{2}{\beta}\int_0^\beta d\tau_1 d\tau_2 \frac{J}{V^2}\sum_{k,k'} G_k\left(\tau_2-\tau_1\right)G_{k'}\left(\tau_2-\tau_1\right)\Big)-1\Big]\sum_m |\Delta_m|^2$$
$$+\Big(\frac{2}{\beta}\int_0^\beta d\tau_1 d\tau_2 \frac{J}{V^2}\sum_{k,k'} G_k(\tau_1-\tau_2)e^{-ik_xa+ik_ya}G_{k'}(\tau_1-\tau_2)\Big)\sum_{<m,n>}\Delta_m\Delta_n cos(\phi_m-\phi_n))$$
$$+\frac{1}{4\beta}\frac{J^2}{V^4}\int_0^\beta d\tau_1 d\tau_2 d\tau_3 d\tau_4 \sum_{k,k',k'',k'''} G_k(\tau_1-\tau_2)G_{k'}(\tau_1-\tau_2)G_{k''}(\tau_3-\tau_4)G_{k'''}(\tau_3-\tau_4)\sum_m |\Delta_m|^4 \tag{4.87}$$

Comparing this to the phenomenological form of[3] as stated in the 3.2, we get:

$$A = \Big[-1+\Big(\frac{2}{\beta}\int_0^\beta d\tau_1 d\tau_2 \frac{J}{V^2}\sum_{k,k'} G_k\left(\tau_2-\tau_1\right)G_{k'}\left(\tau_2-\tau_1\right)\Big)\Big] \tag{4.88}$$
$$C = \Big(\frac{2}{\beta}\int_0^\beta d\tau_1 d\tau_2 \frac{J}{V^2}\sum_{k,k'} G_k(\tau_1-\tau_2)e^{-ik_xa+ik_ya}G_{k'}(\tau_1-\tau_2)\Big) \tag{4.89}$$
$$B = \frac{1}{2\beta}\frac{J^2}{V^4}\int_0^\beta d\tau_1 d\tau_2 d\tau_3 d\tau_4 \sum_{k,k',k'',k'''} G_k(\tau_1-\tau_2)G_{k'}(\tau_1-\tau_2)G_{k''}(\tau_3-\tau_4)G_{k'''}(\tau_3-\tau_4) \tag{4.90}$$

Fourier Transform to Matsubara frequency domain:

$$\boxed{\begin{aligned}
A &= \Big[-1+\Big(\frac{2}{\beta^2}\frac{J}{V^2}\sum_{\omega_n}\sum_{k,k'} G_k\left(i\omega_n\right)G_{k'}\left(-i\omega_n\right)\Big)\Big] && (4.91)\\
C &= \Big(\frac{2}{\beta^2}\frac{J}{V^2}\sum_{\omega_n}\sum_{k,k'} G_k(i\omega_n)e^{-ik_xa+ik_ya}G_{k'}(-i\omega_n)\Big) && (4.92)\\
B &= \frac{1}{2\beta^3}\frac{J^2}{V^4}\sum_{\omega_n,\omega_n'}\sum_{k,k',k'',k'''} G_k(i\omega_n)G_{k'}(-i\omega_n)G_{k''}(i\omega_n')G_{k'''}(-i\omega_n') && (4.93)
\end{aligned}}$$

This is the final expression for the coefficients in Matsubara Frequency and Momentum basis and thereby Free Energy Functional of the form as in [3] is microscopically derived.

4.4 A note on the derived coefficients

On Coefficient A(x,T)

On observing the form of the coefficient, it is clear that at low temperature the value of the $A(x,T)$ is negative since the value of the integral when multiplied by the prefactor is small($\beta = 1/kT$) and as temperature increases this value also increases (since the integral part is directly proportional to temperature) and eventually becomes larger than 1 at some temperature. In that case the value of coefficient A changes from negative to positive and this corresponds to the pseudogap temperature scale on the the phase diagram. In the Free Energy functional, if A is negative it will favour the order parameter magnitude (Δ_m) to be non-zero. This means it will favour the existence of spin-singlet nearest neighbour cooper pairs. However when A becomes positive their existence is not favoured. Moreover if you look at the value of just the integral part without the prefactor (we call this integral GG), we find that this quantity is larger for large doping and the variation w.r.t temperature is minimal. The numerically calculated values of GG can be found in the results section. Then the pseudogap temperature $T^*(x)$ is smaller for large doping. Thereby qualitatively the form of the derived coefficient $A(x,T)$ matches the expectations. The relevant graphs of A(x,T) can be found in the results section.

On Coefficient C(x,T)

First observation about the coefficient $C(x,T)$ is that it is always positive. The integral part without the prefactor (we call it G_kG) is also positive and has the same trend as GG i.e., increases on increasing doping and minimal variation with temperature. Thereby at a particular temperature. The value of C increases with increasing doping. This matches the form of [3] but there is temperature dependence on C as well. C being positive promotes d-wave order since in that case $\cos(\phi_m - \phi_n)$ will prefer to have maximum negative value to minimize free energy and thiz is achieved when there is π-phase difference which is essentially the d-wave OP configuration. Thereby qualitatively the form of the derived coefficient $C(x,T)$ matches the expectations and promotes d-wave superconductivity.

On Coefficient B(x,T)

The coefficient $B(x,T)$ is positive (in order to ensure stabilility), this is essentially the only requirement we have on the coefficient and thereby the qualitatively derived coefficient matches the expectation.

4.5 Manipulating coefficients to a numerically calculable form

Now we discuss possible manipulations of these expressions to get expressions of A, B, C such that the local self energy and the local greens function obtained in real frequency($G^R(\omega)$and $\sum^R(\omega)$) from the self consistency loop developed in [2] can be used to numerically calculate them for different dopings and temperatures.

Method I

We have self energy and local greens function in real frequency thereby we use the spectral representation which serves as the bridge between the real and matsubara frequency quantities.

Using the spectral representation of the Matsubara G.F.

$$G_k(i\omega_n) = \int_{-\infty}^{\infty} d\omega_1 \frac{\rho_G(k,\omega_1)}{i\omega_n - \omega_1} \tag{4.94}$$

Then

$$\sum_{\omega_n}\sum_{k,k'} G_k\left(i\omega_n\right) G_{k'}\left(-i\omega_n\right) = \sum_{\omega_n}\sum_{k,k'} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d\omega_1 d\omega_2 \frac{\rho_G(k,\omega_1)}{i\omega_n - \omega_1}\frac{\rho_G(k',\omega_2)}{-i\omega_n - \omega_2} \tag{4.95}$$

$$\sum_{\omega_n} \frac{1}{i\omega_n - \omega_1}\cdot\frac{1}{-i\omega_n - \omega_2} = \sum_{\omega_n} \frac{-1}{\omega_2 + \omega_1}\Big(\frac{1}{i\omega_n - \omega_1} - \frac{1}{i\omega_n + \omega_2}\Big) = \frac{\beta}{\omega_2 + \omega_1}.(n_F(-\omega_2) - n_F(\omega_1)) \tag{4.96}$$

with the identity $\sum_{\omega_n} \frac{1}{i\omega_n - \omega_1} = \beta n_F(\omega_1)$ where $n_F(\omega)$ is the fermi distribution function.

Using this, the expressions of the coefficients become:

$$A = \Big[\Big(\frac{2}{\beta}\frac{J}{V^2}\sum_{k,k'}\int_{-\infty}^{\infty}\frac{d\omega_1 d\omega_2}{\omega_2+\omega_1}\rho_G(k,\omega_1)\rho_G(k',\omega_2)[n_F(-\omega_2)-n_F(\omega_1)]\Big)-1\Big] \tag{4.97}$$

$$C = \Big(\frac{2}{\beta}\frac{J}{V^2}\sum_{k,k'}e^{-ik_xa+ik_ya}\int_{-\infty}^{\infty}\frac{d\omega_1 d\omega_2}{\omega_2+\omega_1}\rho_G(k,\omega_1)\rho_G(k',\omega_2)[n_F(-\omega_2)-n_F(\omega_1)]\Big) \tag{4.98}$$

$$B = \frac{1}{2\beta}\frac{J^2}{V^4}\sum_{k,k',k'',k'''}\int_{-\infty}^{\infty}\frac{d\omega_1 d\omega_2 d\omega_3 d\omega_4}{(\omega_2+\omega_1)(\omega_3+\omega_4)}\rho_G(k,\omega_1)\rho_G(k',\omega_2)\rho_G(k'',\omega_3)\rho_G(k''',\omega_4) \tag{4.99}$$

$$\times [n_F(-\omega_2)-n_F(\omega_1)][n_F(-\omega_4)-n_F(\omega_3)] \tag{4.100}$$

Here,

$$\rho_G(k,\omega) = -\frac{1}{\pi}Im(G^R(k,\omega)) \quad and \quad G^R(k,\omega) = \frac{1}{\omega+\mu-\epsilon_k-\sum^R(\omega)} \tag{4.101}$$

Here $\sum^R(\omega)$ can be calculated from the Self Consistency loop of [2] and fed into the integral.

In this form the calculations are numerically expensive since it involves 6 integrals in the final form (2 for each k and 2 for ω)

But we observe that

$$\rho_{G_{loc}} = \frac{1}{V}\sum_k \rho_{G_k} \quad and \quad \rho_{G_{loc}} = -\frac{1}{\pi}Im(G^R(\omega))$$

and $\rho_{G_{loc}}$ and $G^R(\omega)$ can directly be accessed from the self-consistency loop.

Using this,

$$A = \Bigg[-1 + \left(\frac{2J}{\beta} \int_{-\infty}^{\infty} \frac{d\omega_1\, d\omega_2}{\omega_1 + \omega_2}\, \rho_{G_{\text{loc}}}(\omega_1)\, \rho_{G_{\text{loc}}}(\omega_2)\, [n_F(-\omega_2) - n_F(\omega_1)] \right) \Bigg]$$

$$C = \left[\frac{2J}{\beta} \int_{-\infty}^{\infty} \frac{d\omega_1\, d\omega_2}{\omega_1 + \omega_2} \left(\frac{1}{V} \sum_k e^{-ik_x a + ik_y a} \rho_G(k, \omega_1) \right) \rho_{G_{\text{loc}}}(\omega_2)\, [n_F(-\omega_2) - n_F(\omega_1)]\, , \right]$$

$$B = \Bigg[\frac{J^2}{2\beta} \int_{-\infty}^{\infty} \frac{d\omega_1\, d\omega_2\, d\omega_3\, d\omega_4}{(\omega_1 + \omega_2)(\omega_3 + \omega_4)} \rho_{G_{\text{loc}}}(\omega_1)\, \rho_{G_{\text{loc}}}(\omega_2)\, \rho_{G_{\text{loc}}}(\omega_3)\, \rho_{G_{\text{loc}}}(\omega_4) \times [n_F(-\omega_2) - n_F(\omega_1)]\, [n_F(-\omega_4) - n_F(\omega_3)] \Bigg] \quad (4.102)$$

This seems like the most appropriate method to calculate the final value since minimal number of integrals are involved and this will only be used to carry out the numerical calculation.

Method 2

We can use the fact in the Matsubara form as stated in 4.91

$$\sum_k f(\epsilon_k) = \int_{-\infty}^{+\infty} d\epsilon D(\epsilon) f(\epsilon) \quad and \quad f(\epsilon_k) = \frac{1}{i\omega_n + \mu - \epsilon_k - \sum(i\omega_n)} \quad (4.103)$$

Using this the form becomes, at least for just A right now:

$$A = \Big[\Big(\frac{2}{\beta^2} \frac{J}{V^2} \sum_{\omega_n} \int_{-\infty}^{+\infty} d\epsilon_1 d\epsilon_2 D(\epsilon_1) D(\epsilon_2) \frac{1}{i\omega_n + \mu - \epsilon_1 - \sum(i\omega_n)} \cdot \frac{1}{-i\omega_n + \mu - \epsilon_2 - \sum(-i\omega_n)} \Big) - 1 \Big] \quad (4.104)$$

$$A = \Big[\Big(\frac{2}{\beta^2} \frac{J}{V^2} \sum_{\omega_n} \int_{-\infty}^{+\infty} d\epsilon_1 d\epsilon_2 D(\epsilon_1) D(\epsilon_2) \sum_{\omega_n} \frac{1}{i\omega_n + \mu - \epsilon_1 - \sum(i\omega_n)} \cdot \frac{1}{-i\omega_n + \mu - \epsilon_2 - \sum(-i\omega_n)} \Big) - 1 \Big] \quad (4.105)$$

Now $D(\epsilon)$ can be taken to be semicircular(bethe lattice density of states) in $d = \infty$

limit.

The expression can be further simplified Here only 3 integrals are involved. But then we have to convert the self energy from the real frequency to Matsubara frequency(Hilbert Transform)[7]

Method III

This is essentially an extension of method 2 only. It relies on identifying the local Green's Functions and using the identities for the Bethe lattice in the infinite dimensional limit.

$$G_{loc}(i\omega_n) = \frac{1}{V}\sum_k G_k(i\omega_n) = \int_{-\infty}^{\infty} d\epsilon_1 \frac{D(\epsilon_1)}{i\omega_n + \mu - \epsilon_1 - \sum(i\omega_n)} \tag{4.106}$$

when we use the Bethe lattice Density of States $\frac{2}{\pi D^2}\sqrt{D^2 - \epsilon^2}$, it becomes a well established identity of contour integrals.

$$\int_{-\infty}^{\infty} d\epsilon_1 \frac{\frac{2}{\pi D^2}\sqrt{D^2 - \epsilon^2}}{z - \sum(i\omega_n)} = \frac{2}{D^2}(z - \sqrt{z^2 - D^2}) \quad where \quad z = i\omega_n + \mu - \sum(i\omega_n) \tag{4.107}$$

Using this the following self consistent equation is obtained:

$$G_{loc}(i\omega_n) = \frac{1}{i\omega_n + \mu - \sum(i\omega_n) - \frac{D^2}{4}G_{loc}(i\omega_n)} \tag{4.108}$$

The local GF obtained through self consistency loop is easier to execute numerically than carrying out the integrals. The inputs to the self-consistency loop would be the $\sum(i\omega_n), \mu$ and it will be found for each $i\omega_n$. In this approach we still need to use a Hilbert Transform to convert $\sum(\omega) \rightarrow \sum(i\omega_n)$

4.6 Dimensional Analysis of Coefficients

If we look at the form in 4.61 and 4.87 we can see that A, C should be dimensionless. Since the term $\sum_m |\Delta_m|^2$ has the dimensions of Energy $[E]$ can be confirmed from first part in 4.87 $\rightarrow e^{-\beta \sum_m |\Delta_m|^2}$

Now for A in equation 4.91, 2 Greens Functions give $1/[E]^2$ J gives $[E]$ and $1/\beta^2$ gives $[E]^2$. Also $\sum_{\omega_n} \rightarrow \int d\omega$ also gives $1/[E]$ So,

$$\frac{1}{[E]^2} \cdot [E] \cdot [E]^2 \cdot \frac{1}{[E]} = [E]^0$$

Thereby A, C are dimensionally accurate.

For B,the term $\sum_m |\Delta_m|^4$ has the dimensions of Energy $[E]^2$, So in net I need $1/[E]$ from the coefficient B. 4 Greens Functions give $1/[E]^4$ J^2 gives $[E]^2$ and $1/\beta^3$ gives $[E]^3$. Also $\sum_{\omega_n,\omega_{n'}} \rightarrow \int d\omega_1 d\omega_2$ also gives $1/[E]^2$ Then,

$$\frac{1}{[E]^4} \cdot [E]^2 \cdot [E]^3 \cdot \frac{1}{[E]^2} = \frac{1}{[E]}$$

B is also dimensionally accurate.

Chapter 5

Numerical Results

Throughout the Numerical Analysis, all the energies have been scales w.r.t. $t = 1eV$, t being the hopping parameter. Also the Bethe density of states was taken to have a semicircular density radius of 2t. Thereby after bandwidth normalization, $T = 1$ corresponds to approximately 2500 K.

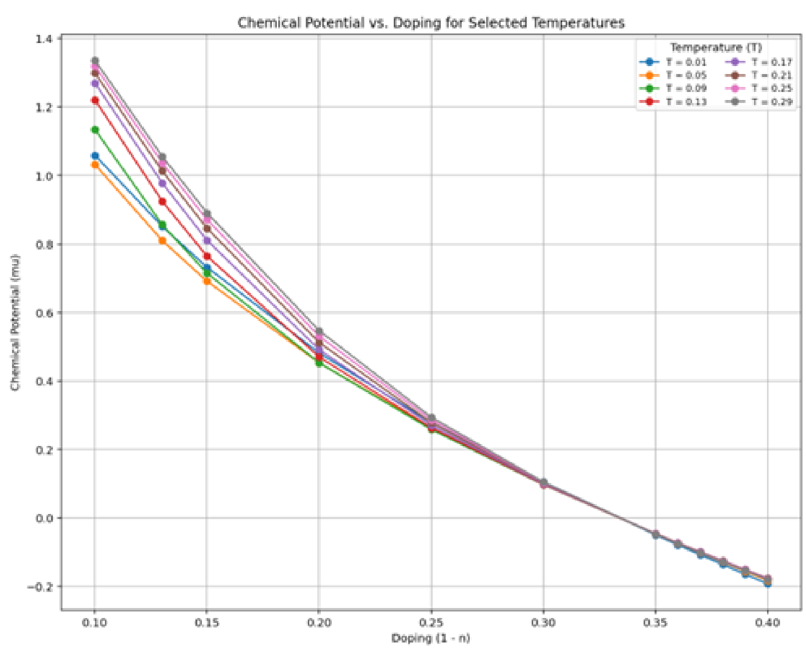
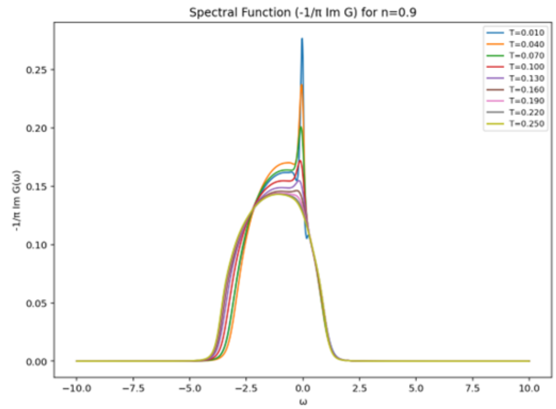
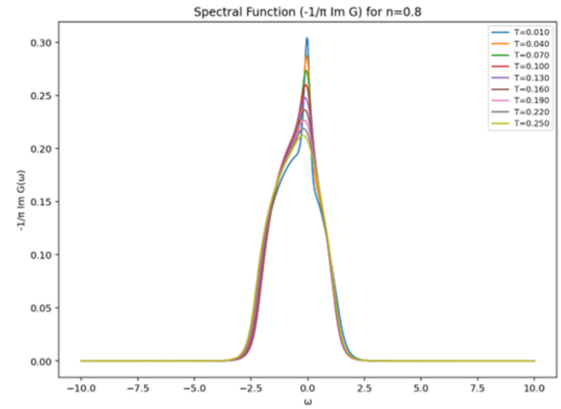


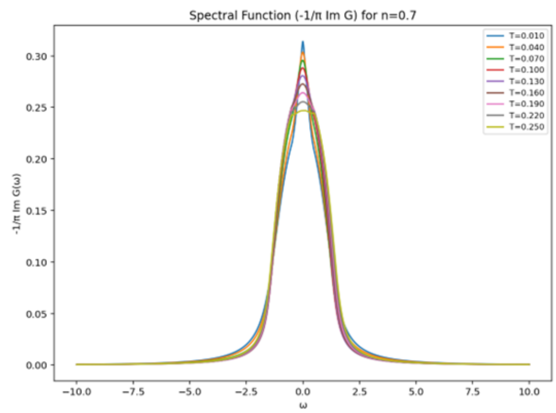
Figure 5.1: Chemical Potential becomes zero at unconventional value (x = 0.33) in X-Operator Formalism



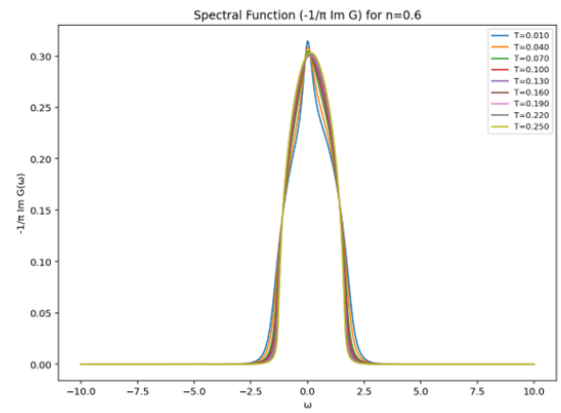
(a) δ= 0.1



(b) δ= 0.2

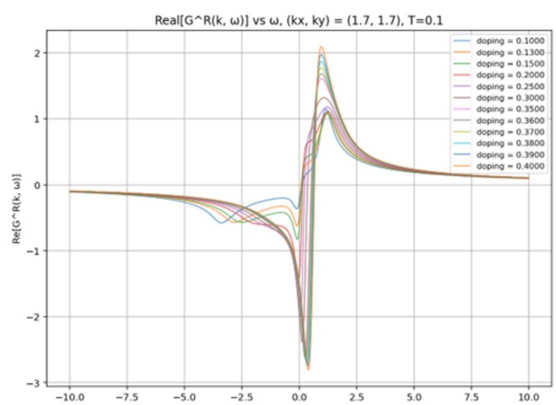


(c) δ= 0.3

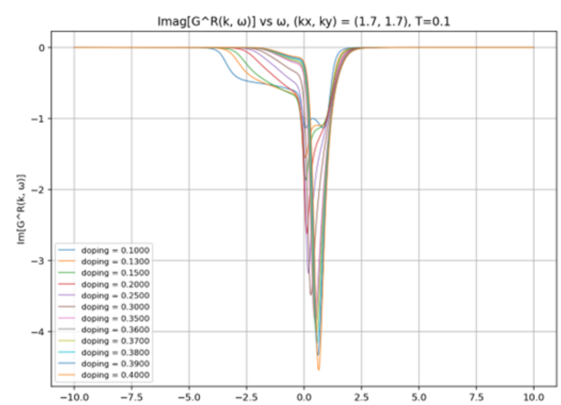


(d) δ= 0.4

Figure 5.2: Spectral Functions-Sharply defined quasiparticle peaks to broad, incoherent spectral weight distributions as temperature increases

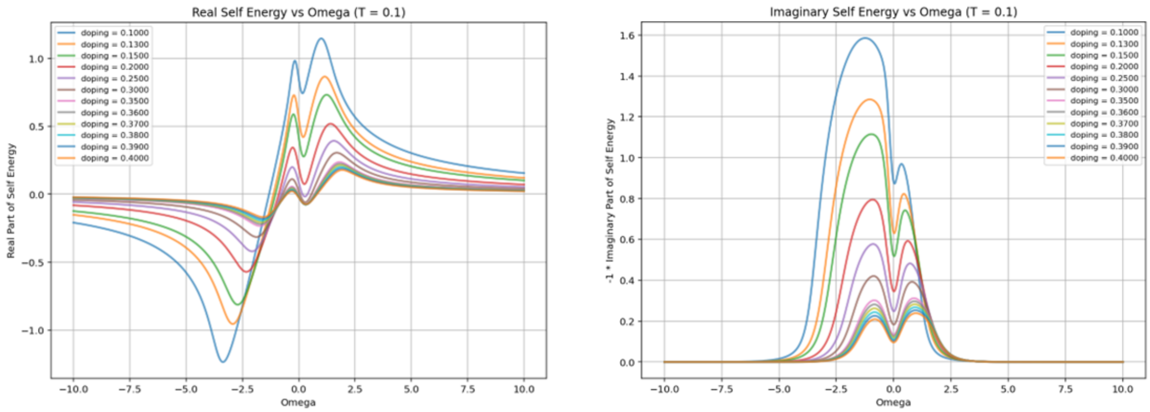


(a) Re($G^R(k, \omega)$)



(b) Im($G^R(k, \omega)$)

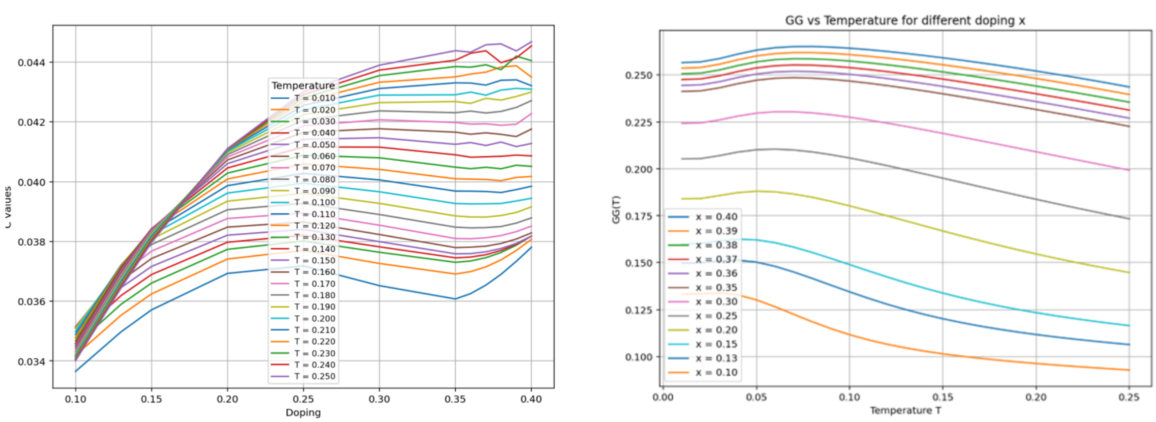
Figure 5.3: Regenerated Momentum Dependent Greens Function $G^R(k, \omega)$



(a) Re($\sum^R(\omega)$)

(b) Im($\sum^R(\omega)$)

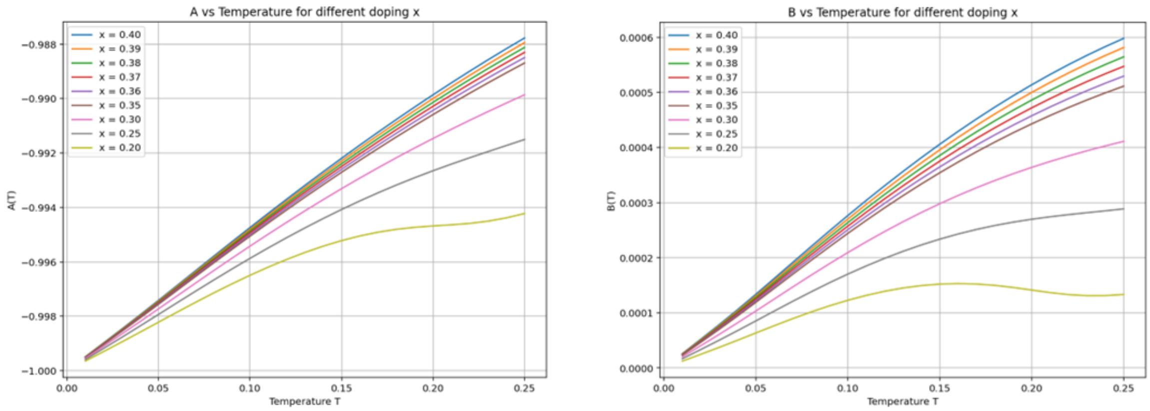
Figure 5.4: Self Energy $\sum^R(\omega)$



(a) Integral of C (GkG)

(b) Integral of A (GG)

Figure 5.5: Integrals involved in microscopic coefficients A, C without the prefactor($2J/\beta$) in 4.102



(a) $A(x, T)$ **(b)** $B(x, T)$

Figure 5.6: Numerically calculated coefficients A and B

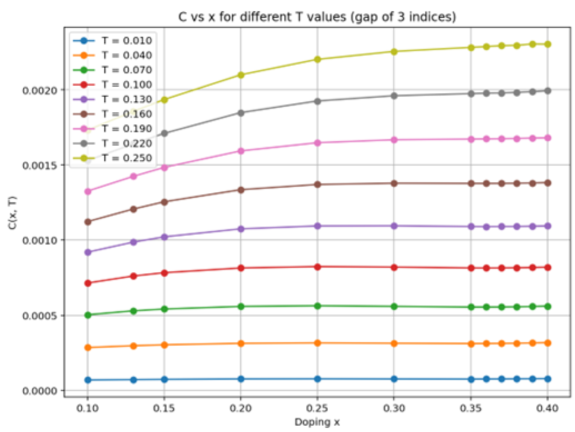
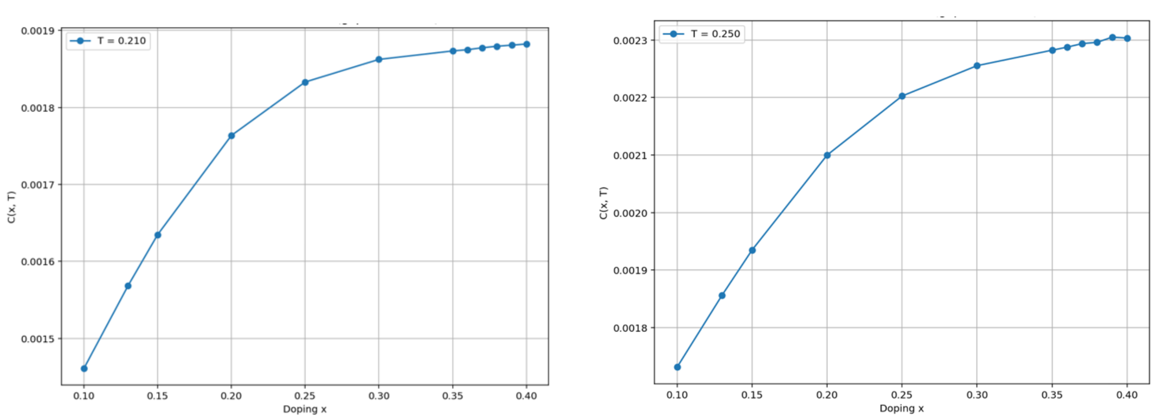


Figure 5.7: Numerically calculated coefficient C



(a) $C(x, T)$ at T= 0.21 **(b)** $C(x, T)$ at T= 0.25

Figure 5.8: Linear Trends in C (matches the phenomenological Expectations)

Superconducting Transition Temperature T_c

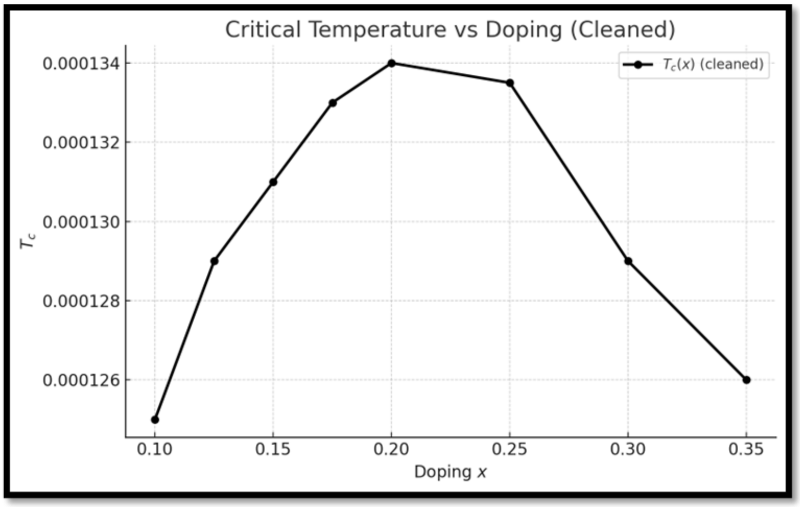


Figure 5.9: Superconducting Dome generated from the Microscopically derived GL coefficients

Chapter 6

Self-Consistency Codes

6.1 Self-Consistency Code to find the Self Energy ($\sum(\omega)$)

6.1.1 Bethe lattice Parameter Calculation Functions

This program contains all the functions that calculates the various parameters for the Bethe lattice which is a well established method for$d = \infty$ limit. The list of expressions that were used in making the loop are summarized in [2] on Pg. 6. This function will be called to the self consistency loop code

```
import numpy as np
import time
from scipy import optimize
from scipy.signal import hilbert
from scipy.interpolate import CubicSpline
import matplotlib.pyplot as plt

def energy_grid (n_g, w = 2):

    return np.linspace (-w, w, n_g)

fermi_function = lambda frequency , beta : 1 / (np.exp(beta * frequency)
     + 1)

bose_function = lambda frequency , beta : 1 / (np.exp(beta * frequency)
    - 1)

def bethe_dos (frequency, w = 2):

    dos = np.sqrt ( w**2 - frequency**2 )
```

```
    dos [np.isnan(dos)] = 0

    return 2*dos/(np.pi * w**2)

def transport_dos (frequency , w = 2):

    trans_dos = np.sqrt (w**2 - frequency**2)
    trans_dos [np.isnan(trans_dos)] = 0

    return (8/(3 * np.pi * w**4))* trans_dos**3

def Hilbert_transform (Z, w = 2):

    ht = np.sqrt ( Z**2 - w**2)

    ht = np.sign ( np.imag (ht)) * ht

    return 2 * (Z - ht )/ (w**2)

def local_greens_function (omega , eta , mu , num , sigma ):

    Q = 1 - num/2
    zeta = (omega + mu + eta)/Q

    return Hilbert_transform ( zeta - sigma )

def greens_function (omega , eta , mu , num , sigma , en):

    Q = 1 - num/2

    zeta = (omega + mu + eta)/Q

    return 1/ (zeta - en - sigma)

spectral_function = lambda gf : (-1/np.pi) * np.imag(gf)

def non_interacting_number_equation (frequency , mu , num , beta):

    Q_1 = 1 - num / 2

    f_1 = (num / 2) / Q_1
```

```
    f_2 = bethe_dos (frequency) * fermi_function (Q_1 * frequency - mu ,
         beta)

    f_2 = np.trapz ( f_2 , frequency )

    return f_1 - f_2

def interacting_number_equation (omega , eta , mu , num , sigma , beta):

    nd = spectral_function (local_greens_function (omega , eta , mu ,
        num , sigma)) * fermi_function (omega , beta)

    return num - (2 * np.trapz (nd , omega))


def current_current_correlation (omega , energy , eta , mu , num , sigma
    , beta):

    n_f = fermi_function (omega , beta)

    phi_epsilon = transport_dos (energy)

    Xi = 0

    en_length = len(energy)

    for i_1 in range (en_length):

        A_eps = spectral_function (greens_function(omega , eta , mu ,
            num , sigma , energy[i_1]))

        T_1 = np.fft.fft (A_eps * n_f ) * np.fft.ifft (A_eps)

        T_2 = np.fft.fft (A_eps) * np.fft.ifft (A_eps * n_f)

        Xi += phi_epsilon [i_1] * (T_1 - T_2)


    im_chi = np.fft.fftshift (np.fft.fft(Xi)) * abs (omega[1] - omega
        [0])
```

```
    re_chi = - np.imag (hilbert(np.real (im_chi), 4*len(im_chi))[:len(
        im_chi)])

    return  - 2*np.pi * (re_chi + 1j* im_chi) * abs (energy[1] - energy
        [0])


def grt_curr_img_zero (omega , energy , eta , mu , num , sigma  , beta):

    df = 1 / (1 + np.cosh(beta*omega))

    en_length = len(energy)

    phi = transport_dos(energy)

    img_zero = 0

    for i in range (en_length):

        A_eps = spectral_function (greens_function(omega , eta , mu ,
            num , sigma , energy[i]))

        img_zero += A_eps** 2 * phi[i]

    img_zero = np.trapz (img_zero * df, omega)

    return - np.pi * img_zero * abs(energy[1] - energy[0])


def charge_spin_correlation (omega ,  num , curr_corr, gc_zero_freq,
    beta):

    img_grt_curr = np.imag(curr_corr)

    img_grt_curr = ( 1 + bose_function(omega , beta)) * img_grt_curr

    img_grt_curr [len(omega)//2] =  gc_zero_freq

    re_grt_curr = - np.imag (hilbert(img_grt_curr, 4*len(omega))[:len(
        omega)])

    grt_curr = re_grt_curr + 1j * img_grt_curr
```

```
    D_c_1 = -num / (omega + grt_curr/num)

    D_s_1 = -(0.5*num) / (omega + grt_curr/(0.5*num))


    #return grt_curr

    return (-0.25 * D_c_1 - 1.5 * D_s_1)

def self_energy (omega , mu , num, sigma , Bosonic_gf , Beta, eta):


    Q = 1 - num / 2

    bose_sp_fun = spectral_function (Bosonic_gf)

    hyperbolic_fun_1 = np.tanh (Beta * omega/2)

    hyperbolic_fun_2 = 1 / hyperbolic_fun_1

    hyperbolic_fun_2 [np.isinf (hyperbolic_fun_2) ] = 1e6

    rhod = bose_sp_fun * hyperbolic_fun_2

    omega_1 = omega[len(omega)//2 + 1:]

    omega_2 = omega[0 : len(omega)//2]

    cs_1 = CubicSpline (omega_1 , rhod[len(omega)//2 + 1 :])

    cs_2 = CubicSpline (omega_2 , rhod[0:len(omega)//2])

    rhod[len(omega)//2] = (cs_1(0) + cs_2(0)) * 0.5

    fermion_sp_fun = spectral_function (local_greens_function (omega ,
        eta ,  mu , num , sigma))

    T_1 = np.fft.fft ( fermion_sp_fun * hyperbolic_fun_1)  * np.fft.fft
        (bose_sp_fun)

    T_2 = np.fft.fft ( fermion_sp_fun) * np.fft.fft (rhod)
```

```
    SE = np.fft.fftshift(np.fft.ifft( T_1 + T_2)) * abs(omega[1] - omega
        [0])

    SE_img = (-(np.pi)/(2*Q**2)) * SE

    SE_re = - np.imag (hilbert(np.real (SE_img), 4*len(SE_img))[:len(
        SE_img)])

    return (SE_re + 1j * SE_img)
```

6.1.2 Self-Consistency Loop

This programs calls the functions from the last program according to the loop discussed in [2] and give the values of $\sum(\omega)$ for different doping values. It uses Hieracrchal Data Files(HSF5) to store the data and using Multi-core processing as well.

```
from bethe_symmetric_charge import *
import h5py
import multiprocessing as mp
import os


def self_consistency_loop (n, beta , iter_max = 50):

    np.seterr(divide = 'ignore', invalid = 'ignore')

    Omega = np.arange (-10,10,20/8192)

    energy = energy_grid (1001)

    eta = 2j * abs (Omega[1] - Omega[0])

    se = 0

    mu = optimize.root_scalar (lambda x :
        non_interacting_number_equation (energy , x , n , beta),
                             x0 = -2 , x1 = 2 ).root


    for j in range (iter_max):
```

```
curr = current_current_correlation (Omega , energy , eta , mu ,
    n , se, beta)

gc_zero_freq = grt_curr_img_zero (Omega , energy , eta , mu , n
    , se  , beta)

gc = charge_spin_correlation (Omega ,  n , curr , gc_zero_freq,
    beta)

charge_spectral = spectral_function(gc) * (1 - np.exp(-beta *
    Omega))

#charge_spectral [0:len(Omega)//2] = - np.flip(charge_spectral[
    len(Omega)//2 + 1:])

charge_spectral [len(Omega)//2 - 1: 1 :-1] = - (charge_spectral[
    len(Omega)//2 + 1 :len(Omega) - 1 ])

charge_spectral [0] = 0

charge_spectral [1] = 0

se = self_energy (Omega , mu , n , se , 1j * np.pi *
    charge_spectral , beta , eta)

delta_n = interacting_number_equation (Omega, eta , mu , n , se
    , beta)

if j%10 == 0:

    print(j, delta_n)

if abs(delta_n) < 1e-5:

    with h5py.File(f"{directory}/n_{n}_beta_{beta}.h5",'w') as A
        :

        A.create_dataset('omega',data =  Omega)
        A.create_dataset ('se', data = se)
        A.create_dataset ('local_gf', data =
            local_greens_function (Omega , 0 , mu , n , se))
        A.create_dataset ('charge_spin_spectralfun', data =
```

```
                    charge_spectral)
                A.create_dataset ('dc_conductivity', data = - beta *
                    grt_curr_img_zero (Omega , energy , 0 , mu , n, se ,
                     beta) )
                A.create_dataset ('retarded_current_correlation',data =
                     curr)
                A.create_dataset ('delta_n', data = delta_n)

            A.close()

            break

        mu = optimize.root_scalar (lambda x: interacting_number_equation
             (Omega , eta , x , n , se , beta)
                                  , x0 = -2, x1 = 2).root

# Define the target directory
directory = "D:/Downloads/Self consistent Data"

# Ensure the directory exists (optional but recommended)
os.makedirs(directory, exist_ok=True)


if __name__ == "__main__":

    number = [0.7,0.8,0.9]
    Temperature = np.zeros(20)
    Temperature[0:10] = np.linspace (0.001 ,0.3 , 10)
    Temperature[10:] = np.linspace (0.31, 1 , 10)

    var_list = []

    for i in number:
        for j in Temperature:

            var_list.append ([i,1/j])

    n_cpu = mp.cpu_count()
    pool = mp.Pool (n_cpu - 1)
    pool.starmap (self_consistency_loop , var_list)
    pool.close()
```

Chapter 7

Summary and Outlook

We have successfully calculated the Final Free energy functional for the case of site dependent and time independent auxillary field. The Microscopically derived expressions of coefficients as stated in 4.91 are based out of the fact that $(U/t \to \infty)$ magnitude of the order processes and how physically viable they are.Throughout the calculations we have ensured that there are no conventional Gorkov-like approximations were made. on the order parameter however other approximations like the decoupling approximation and the Long wavelength approximation were essential to bring the coefficients to a single particle Greens function form. On analyzing the Coefficients, it is clear that they give the expected qualitative trends 4.4, however they don't give the expected values at the temperatures we would expect them to. This is because the integral along with the prefactor is a very small quantity and is not able to end the pseudogap phase or onset global coherence at the expected temperatures.

The most obvious emergence of this anomaly seems to be the decoupling approximation, where we replace the 4 point correlator by a 2 point correlator. But we are limited to calculating the single particle retarded greens function by the self-consistency loop. Better methods to decouple the correlator is needed, for instance [53]. However coming up with a decoupling formalism of such nature for our Hamiltonian and that too in HUbbard operators is a complicated Task and will essentialy be the next meaningful pursuit of this work.Moreover the self-consitency loop itself[2] has its own limitation both numerical (in its current form) and analytical(no vertex corrections are considered for charge correlator). Another aim of further work on this topic would be deal with these shortcomings.

Even in the present form, coming up with a better decoupling approximation and capturing two temperature scales ($T^*(x)$ and $T_c(x)$) in a theory of microscopic origins would be a major advancement towards understanding Strongly Correlated Electron Systems!!

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Appendix A

Discussion on the approximations

A.1 Long Wavelength approximation

It is absolutely essential to get the final GL Equation to apply a long wavelength approximation. Although initial attempts were made to prove that this limit is sensible and works in the $d = \infty$ limit but actually it could not be proved because the scaling of the 4-point correlator in X operator formalism is unknown. The coarse graining of the fields is an integral part of deriving the GL theories. However due to the short coherence length ξ in cuprates, a generic GL coarse-graining is not valid. To solve this issue we do a minimal unconventional coarse graining of the functional where site lattice is replaced by bond lattice which we say is a relatively coarser lattice. This is enough to take care of the ultra-fast fluctuations that made the raw operator meaningless at a single site.

Thereby taking just the local term and the nearest neighbour term in the bond lattice representation is enough. The nearest neighbor term is necessary for phase coherence and we claim that just these 2 terms are enough to capture the essential physics. The approach has its own limitations specially in the pseudopgap phase and in the highly underdoped and overdoped region near the quantum critical point where phase fluctations are important. In the table below I give a detailed arguments on the validity of the approximation in various regions of phase diagram.

A.2 Translationally Invariant Retarded Greens Functions

Taking Greens function to be translation invariant which is essentially true since there are no disorders in the system and the doping is considered to be uniform throughout the lattice. This is a meaningful approximation since we are dealing with uniform doped lattices

Region of the phase diagram	Validity	Reasoning
Over-doped & much of optimal (inside the SC dome)	**Works**	Cooper pairs extend over *several* Cu–O bonds, so the order-parameter field changes gently; the NN term already captures the local slope and farther couplings are negligible.
Moderately under-doped & just above T_c	**Partially works**	Pair size is only a *few* bonds. The smooth picture is still reasonable for low fields and long-wavelength probes, but next-neighbour and phase-fluctuation terms begin to affect quantitative results.
Deep under-doped / pseudogap region	**Fails**	Pairs span merely one–two bonds; the order parameter can vary bond-to-bond, so omitting farther couplings ignores essential non-local physics.
Strong magnetic fields or high-frequency (short-λ) probes	**Fails**	Vortex cores or light wavelengths approach the pair size, injecting large-q components beyond the long-wavelength window; a non-local kernel is required.
Far above T_c (normal / fluctuating state)	**Fails**	Superconducting correlations are very short-ranged; a smooth GL order parameter no longer exists, so the approximation is inapplicable.

Table A.1: Validity of long-wavelength approximation for various phases of cuprates

and is essential to achieve to achieve a Free Energy Functional of the desired form. Mathematically this means that:

$$G_{ij}(\tau_1 - \tau_2) = G(i - j, \tau_1 - \tau_2)$$

Along with this we also use the $d = \infty$ limit and in the spirit of DMFT[54] regenrate :

$$G_k(\omega) = \frac{1}{\omega - \epsilon_k + \mu - \sum(\omega)}$$

This form satisifies the follwing proprty for 2D square lattice:

$$\boxed{G_{k_x,k_y} = G_{k_x,-k_y} = G_{-k_x,k_y} = G_{-k_x,-k_y}}$$

A.3 Decoupling approximation

This approximation is essentially heuristic and was necessary because our inability to calculate 4 point Greens Functions in the existing framework: The decoupling approximation replaces every four–operator correlator of projected Hubbard (or $t-J$) X–operators by a product of two *local* single–particle Green functions,

$$\langle X_i^{\alpha 0} X_j^{0\beta} X_j^{\beta 0} X_i^{0\alpha} \rangle \implies G_{ii}^{\alpha}(\tau)\, G_{jj}^{\beta}(\tau),$$

thereby discarding all connected four-point cumulants. This is the leading $(1/Z^0)$ term of a $1/Z$ expansion ($d \to \infty$ or DMFT limit).

- **Keeps:** on–site amplitude physics, static charge/spin renormalisation; quantitatively accurate where correlations are local (over-doped Fermi liquid, high-T strange metal).
- **Ignores:** every non-local or dynamic vertex $\Rightarrow$ no Goldstone/vortex phase fluctuations, no long-range AF ladders, no quantum-critical scaling.

Region (x, T)	**Validity**	**Reasoning**
Over-doped FL ($0.18 \lesssim x \lesssim 0.30,\ T \ll T_F$)	Reliable	Correlations local; quasiparticles well-defined.
Strange metal above T_c ($0.12 \lesssim x \lesssim 0.18,\ T_c < T \lesssim 2J$)	Semi-quantitative	Thermal smearing reduces vertices, but AF/-pair fluctuations start to matter.
Inside SC dome ($T < T_c(x)$)	Amplitude OK Phase missing	Gap size captured; lack of Goldstone and vortex physics distorts ρ_s and $\lambda(T)$.
Pseudogap regime ($x \lesssim 0.12,\ T_c < T < T^{(x))}$	Fails	RVB strings and AF ladders as large as bubble term; pseudogap underestimated.
Mott / AF edge ($x \lesssim 0.05$)	Fails	Long-range AF and constraint physics violate Gaussian closure.
Quantum-critical tail ($x \gtrsim 0.30,\ T \to 0$)	Needs charging term	Weak pairing; order lost through quantum phase fluctuations absent in this action.

Table A.2: Validity of decoupling approximation for various phases of cuprates

Appendix B

Parity Anomaly in the $t - J$ Model

In the derivation of the Free Energy Functional, we performs a Hubbard–Stratonovich (HS) transformation to decouple the four-fermion spin exchange term. This introduces auxiliary bosonic fields which mediate interactions and simplify the fermionic part of the action. Then we study the resulting free energy (or effective action) of these auxiliary fields after integrating out the fermions.

In (2+1) dimensions, however, integrating out massless fermions coupled to gauge-like auxiliary fields introduces a subtle quantum effect known as the *parity anomaly*[51, 52].

The *t*-J model is given by:

$$H_{tJ} = -t \sum_{\langle ij \rangle, \sigma} \left(\tilde{c}_{i\sigma}^{\dagger} \tilde{c}_{j\sigma} + \text{h.c.} \right) + J \sum_{\langle ij \rangle} \left(\vec{S}_i \cdot \vec{S}_j - \frac{1}{4} n_i n_j \right) \tag{B.1}$$

Here, $\tilde{c}_{i\sigma}$ are projected electron operators enforcing the no-double-occupancy constraint. A complex bond field $\chi_{ij} \sim \langle \tilde{c}_i^{\dagger} \tilde{c}_j \rangle$ is introduced through an HS transformation:

$$\vec{S}_i \cdot \vec{S}_j \to \chi_{ij} \tilde{c}_j^{\dagger} \tilde{c}_i + \text{h.c.} + \frac{|\chi_{ij}|^2}{J} \tag{B.2}$$

In mean-field configurations such as the π-flux phase, the bond fields χ_{ij} acquire phases that generate a π flux per plaquette, producing a Dirac-like dispersion for the fermions at low energy.

More precisely, in the π-flux phase, expanding the fermion dispersion near the nodal points $(\pm\pi/2, \pm\pi/2)$ reveals that the low-energy excitations are described by massless Dirac fermions:

$$H_{\text{eff}} = \bar{\psi}(x)\, i\gamma^{\mu} \partial_{\mu} \psi(x) \tag{B.3}$$

Phase fluctuations of the bond fields, modeled as $\chi_{ij} = |\chi| e^{iA_{ij}}$, couple minimally to the fermions and act as an emergent gauge field A_μ. The effective action becomes:

$$S_{\text{eff}}[\psi, A] = \int d^3x\, \bar{\psi}(x)\, i\gamma^\mu(\partial_\mu + iA_\mu)\psi(x) \tag{B.4}$$

To obtain the free energy for the auxiliary field A_μ, we integrate out the massless Dirac fermions:

$$Z[A] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\, e^{-S_{\text{eff}}[\psi,A]} = \det\big(i\not{D}[A]\big) \tag{B.5}$$

This determinant is UV-divergent and requires regularization. In (2+1) dimensions, *any* gauge-invariant regularization, such as Pauli–Villars, necessarily breaks parity and yields a non-zero Chern–Simons term:

$$\log\det\big(i\not{D}[A]\big) \supset \pm\frac{i}{8\pi}\int d^3x\, \epsilon^{\mu\nu\rho}A_\mu\partial_\nu A_\rho \tag{B.6}$$

This term is odd under parity and represents the *parity anomaly*: although the classical action is parity-invariant, the regularized quantum effective action is not.

In our derivation of the Ginzburg–Landau-like effective functional for the auxiliary field (such as the bond field), the following conditions are satisfied:

1. The HS transformation introduces a complex field whose phase behaves like a $U(1)$ gauge field.

2. The mean-field spectrum has Dirac nodes (e.g., in the π-flux phase).

3. The fermionic determinant is required to be gauge-invariant due to the emergent local symmetry.

4. Therefore, the regularization used must preserve gauge invariance and thus breaks parity.

Although a regularization scheme is not explicitly introduced in our derivation of the effective free energy functional, it is implicitly assumed as part of evaluating the fermionic determinant. The path integral over fermions yields a formally divergent determinant, which must be regularized to be well-defined. Since the auxiliary fields introduced via Hubbard–Stratonovich transformation couple to fermions in a gauge-like manner, we require gauge invariance to be preserved in the quantum effective action. In (2+1)D, however, any gauge-invariant regularization — including the one implicitly used here — necessarily breaks parity symmetry. This leads to the emergence of the parity anomaly in our low-energy effective theory, and justifies the sign structure and convergence behavior of the resulting free energy functional

Appendix C

Calculating Transition Temperature $T_c(x)$

Determination of T_c and Related Quantities in the Mean-Field Framework

In the mean-field theory approach to superconductivity, the transition temperature T_c is obtained by analyzing the onset of long-range phase coherence of the complex order parameter $\psi_m = \Delta_m e^{i\phi_m}$. This order parameter is represented by a planar spin $\mathbf{S}_m = (\Delta_m \cos\phi_m, \Delta_m \sin\phi_m)$, and interacts with neighboring spins through a coupling constant C. The relevant ideas are discussed in [3]

The self-consistent mean-field equation relates the effective staggered field $\mathbf{h} = (h_x, h_y)$ acting on each spin to its thermal average:

$$h_\alpha = 4C\langle S_\alpha \rangle_0 \quad (\alpha = x, y), \tag{C.1}$$

where the thermal average is given by

$$\langle S_\alpha \rangle_0 = \left(\frac{h_\alpha}{h}\right) \frac{\int_0^\infty \Delta^2 d\Delta\, P_0(\Delta) I_1\left(\frac{h\Delta}{T}\right)}{\int_0^\infty \Delta\, d\Delta\, P_0(\Delta) I_0\left(\frac{h\Delta}{T}\right)}, \tag{C.2}$$

with $h = |\mathbf{h}|$ and I_0, I_1 being modified Bessel functions. The distribution function $P_0(\Delta)$ is defined as:

$$P_0(\Delta) = \exp\left[-\beta\left(A\Delta^2 + \frac{B}{2}\Delta^4\right)\right]. \tag{C.3}$$

The superconducting transition temperature T_c is determined by the condition at which the system first supports a non-zero value of $\mathbf{h}$. This leads to the criterion:

$$2C \left\langle \Delta^2 \right\rangle_{P_0} \Big|_{T=T_c} = T_c, \tag{C.4}$$

where

$$\left\langle \Delta^2 \right\rangle_{P_0} = \frac{\int_0^\infty \Delta^3 P_0(\Delta)\, d\Delta}{\int_0^\infty \Delta P_0(\Delta)\, d\Delta}. \tag{C.5}$$

In addition to T_c, other physical quantities can be obtained using the self consistent solution of C.1 , the *superfluid stiffness* ρ_s, which characterizes the rigidity of the superconducting phase, is computed using:

$$\rho_s = -\frac{C}{2N_b} \left\langle \sum_{\langle m,m' \rangle} \Delta_m \Delta_{m'} \cos(\phi_m - \phi_{m'}) \right\rangle. \tag{C.6}$$

This expression captures the contribution of phase coherence between neighboring sites, and plays a central role in determining electromagnetic response functions such as the penetration depth.